Foundations of Computer Science

A Complete Theory & Lab Manual

Computer Science Department

2026 Edition

Computers Windows Shortcuts Word Excel PowerPoint Internet Project

0 Preface

Welcome to Your Computer Science Adventure!

Hello there, future tech expert! 🚀

Welcome to **Foundations of Computer Science** — your personal guide to understanding the amazing world of computers. Whether you've been using computers for years or you're just getting started, this book is designed to take you on an exciting journey from the very basics all the way to creating your own mini-projects. Think of it as your friendly companion that explains everything in plain, simple language — no confusing jargon, we promise!

This book is organised into **eight modules**, and each one builds on what you learned before. You'll start by discovering what a computer actually *is* (spoiler: it's not magic, even though it sometimes feels like it!). Then you'll explore Windows, master keyboard shortcuts that will make you faster than your friends, create professional documents in Microsoft Word, crunch numbers in Excel, build stunning presentations in PowerPoint, and learn to navigate the internet safely. By the end, you'll bring everything together in a capstone project that shows off all your new skills.

Every chapter follows a simple pattern: first, we explain the **theory** — the "what" and "why." Then, we give you **shortcuts and tips** to work smarter, not harder. Finally, you get **hands-on lab practicals** where you actually *do* things on a real computer. Oh, and keep an eye out for **Byte** — our friendly robot mascot who pops up throughout the book with tips, fun facts, and the occasional warning. Byte's got your back!

One last thing: **don't worry about getting everything right the first time.** Seriously. Experiment, make mistakes, click the wrong button, accidentally change the font to something ridiculous — and then figure out how to fix it. That's exactly how real computer scientists work. Every expert was once a beginner, and every great discovery started with someone saying, "I wonder what happens if I try this. . . "

Happy learning!

The Computer Science Department

Hi! I'm **Byte**, your robot guide! Whenever you see me, I'll share a cool tip, a fun fact, or a helpful warning. Let's have fun learning together!

0 How to Use This Book

This book is packed with colourful boxes, icons, and special sections to help you learn. Here's a quick guide to what each one means:

Theory Box Learn the concept	**Fun Fact** Cool trivia	**Note** Important info
Warning Be careful!	**Shortcuts** Save time	**Try It** Quick challenge
Byte's Tip Robot advice	**Lab Activity** Hands-on practical	**Challenge** Extra task

Other Symbols You'll See

`Ctrl` **Keyboard keys** are shown as styled key caps like this.

✓ A **check mark** means "Yes" or "Correct."

✗ A **cross mark** means "No" or "Incorrect."

› Points in a **learning objectives** list.

? Marks a **review question** at the end of a chapter.

1 A **step number** in a lab practical.

Note

Every chapter follows this pattern: **Learn** (theory) → **Practice** (shortcuts & tips) → **Do** (lab practicals). Work through them in order for the best experience!

Look for the **QR codes** at the end of each Part — they link to extra resources, videos, and activities chosen by your teacher. Scan them with your phone's camera!

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PART I

Getting Started with Computers

Discover what computers are, how they work, and why they matter in your world.

Hardware Input Output CPU Software

Part I

Getting Started with Computers

1 Introduction to Computers

Learning Objectives

- Define what a computer is and explain its purpose
- List the main characteristics of a computer
- Identify how computers are used in daily life
- Understand the difference between data and information

Welcome to Chapter 1! Get ready to discover what makes computers so amazing. Spoiler alert — they're basically super-fast, super-accurate thinking machines!

1.1 What Is a Computer?

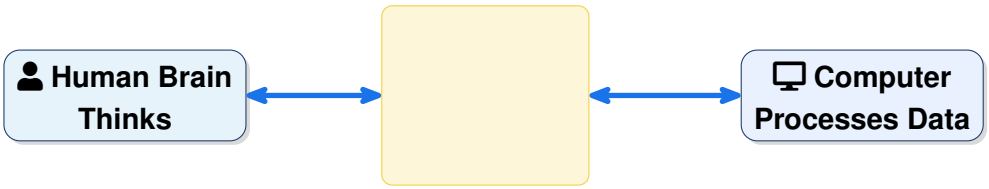
A computer is an electronic device that accepts **data** (raw facts and figures), processes it according to a set of instructions (called a **program**), and produces useful **information** as output.

Definition: Computer

A **computer** is an electronic machine that:

1. **Accepts** data (input)
2. **Processes** data (using programs)
3. **Stores** data (in memory)
4. **Produces** results (output)

Think of it like this: your brain takes in information through your eyes and ears, thinks about it, remembers important things, and then tells your body what to do. A computer works in a very similar way!



Both can do these — but computers are MUCH faster!

Figure 1.1: A computer is like a super-fast brain that processes data

Real World Connection

When you type your name into a school computer, that's **data**. When the computer uses it to print your name on a report card, that's **information**. Data becomes information when it's processed and made useful!

1.2 Characteristics of a Computer

What makes computers special? Here are the six key characteristics that make them so powerful:

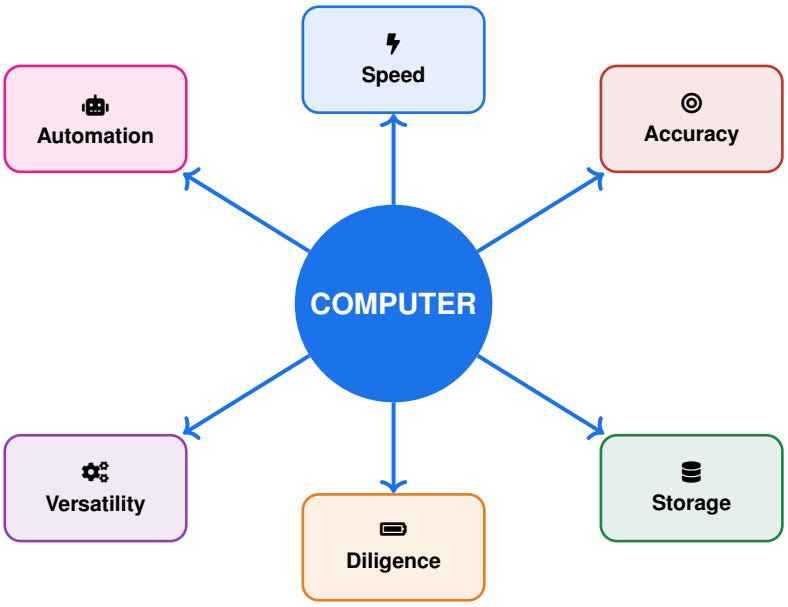


Figure 1.2: The six key characteristics of a computer

1. **Speed** — Computers can perform *billions* of calculations per second. What takes you minutes takes a computer less than a blink!

2. **Accuracy** — Computers almost never make mistakes. If the input data is correct, the output will always be correct.

3. **Storage** — Computers can store enormous amounts of data — from your holiday photos to entire libraries of books.

4. **Diligence** — Unlike humans, computers never get tired or bored. They can work for hours without slowing down.

5. **Versatility** — The same computer can play music, write essays, solve maths problems, and browse the internet.

6. **Automation** — Once programmed, computers can perform tasks automatically without human help.

A modern computer can perform over **1 billion calculations per second**. That means if you could do one calculation per second, it would take you about **31 years** to do what a computer does in **one second**!

1.3 Computers in Daily Life

Computers are everywhere — and we mean *everywhere*! Let's look at where you might encounter them:

Table 1.1: Computers in daily life

	Where	What We Use It For	Example
	School	Writing assignments, research	MS Word, Google
	Home	Entertainment, communication	YouTube, Games
	Bank	Transactions, account management	ATM, Online Banking
	Hospital	Patient records, diagnosis	Medical software
	Shop	Billing & inventory	POS system

Try It Yourself!

Think about your typical day. Can you list **five** different times you (or your family) use a computer or a computerised device? Write them down! Remember, smartphones and smart TVs have computers inside them too.

1.4 Data and Information

Data vs Information

- **Data** = Raw, unprocessed facts. Example: `85, 92, 78, 90, 88`
- **Information** = Processed, meaningful data. Example: "Average marks = 86.6"
- **Processing** = The step where data is converted into information

Did You Know?

The word "computer" originally referred to a **person** who did calculations by hand! It wasn't until the 1940s that the word started to mean an electronic machine. The first electronic computer, **ENIAC** (1945), weighed 27 tonnes and filled an entire room. Today, your smartphone is millions of times more powerful!

Review Questions

Q1. What is a computer? Write a definition in your own words.

Q2. List any four characteristics of a computer.

Q3. Give two examples each of **data** and **information**.

Q4. Name three places where computers are used in daily life.

Q5. Why is a computer said to be "diligent"?

Q6. What is the difference between data and information?

Challenge Task

Interview a family member or neighbour about how they use computers at work. Write a short paragraph (5–6 sentences) about what you learned. Include the **type of computer** they use and at least **two tasks** they do on it.

2 Parts of a Computer

Learning Objectives

- Identify the main physical parts of a computer system
- Distinguish between input devices and output devices
- Explain the difference between hardware and software
- Understand the role of the CPU (processor)

In this chapter, you'll meet all the "body parts" of a computer. Think of it like learning anatomy — but way more fun and with zero exams on bones!

2.1 The Computer System

A computer system is made up of different parts that work together. Just like a human body has a brain, eyes, ears, and hands, a computer has its own special parts.

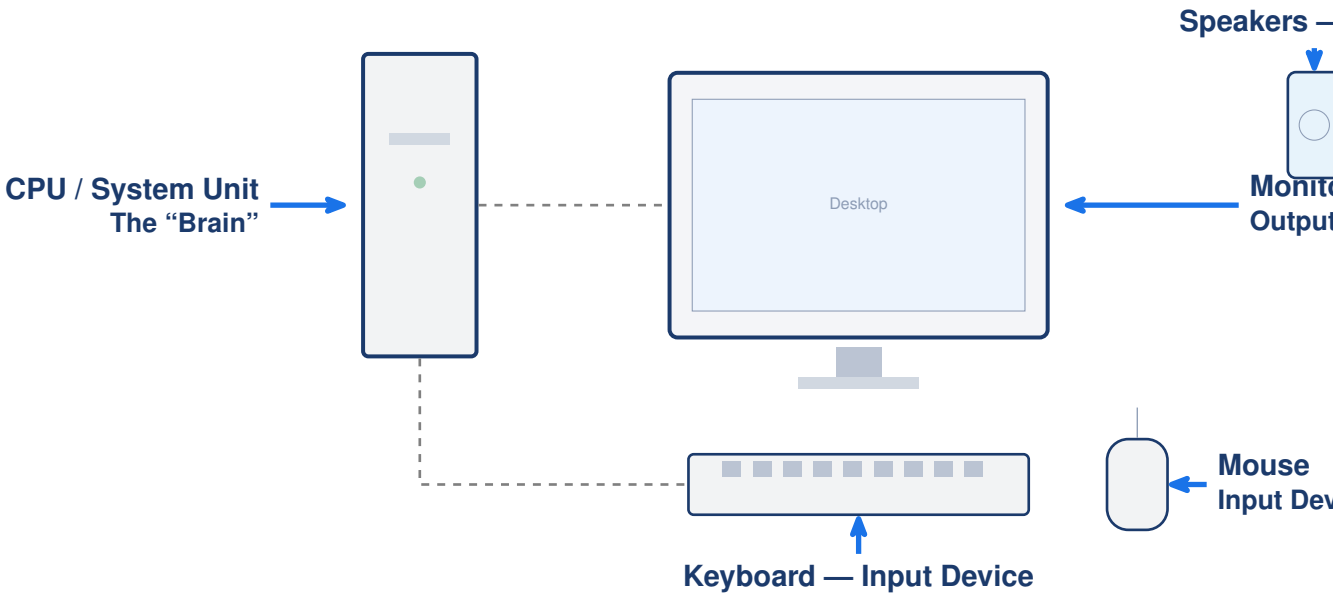


Figure 2.1: A typical desktop computer setup with all parts labelled

2.2 Hardware vs Software

The parts of a computer fall into two big categories: **hardware** and **software**.

Hardware and Software

- **Hardware** = The physical parts you can *touch* (monitor, keyboard, mouse, CPU)
- **Software** = The programs and instructions that tell the hardware what to do (Windows, MS Word, games)

Without hardware, software has nothing to run on. Without software, hardware is just an expensive paperweight!

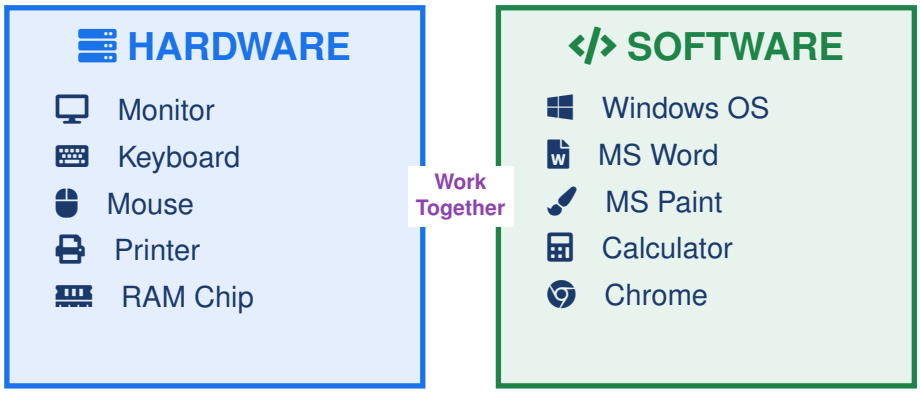


Figure 2.2: Hardware (physical parts) vs Software (programs) — they need each other!

2.3 Input and Output Devices

Input and Output

- **Input devices** send data *into* the computer (e.g., keyboard, mouse)
- **Output devices** show results *from* the computer (e.g., monitor, printer)

Table 2.1: Input devices vs Output devices

Input Devices	Output Devices
Keyboard	Monitor
Mouse	Printer
Scanner	Speakers
Microphone	Projector
Webcam	Headphones

The CPU is like the **brain** of the computer — it does ALL the thinking! Every instruction, every calculation, every decision goes through the CPU. Without it, a computer is just a fancy box.

2.4 The CPU — The Brain

The CPU (Central Processing Unit) is the most important part inside the computer. It processes all the data and instructions. When people say a computer is "fast," they usually mean it has a powerful CPU.

Note

The CPU is a small chip — usually no bigger than a postage stamp — but it contains **billions** of tiny electronic switches called transistors. These switches flip on and off billions of times every second to process data!

Did You Know?

The CPU inside your school's computer is more powerful than all the computers NASA used to send astronauts to the Moon in 1969. That's right — your calculator app has more computing power than the Apollo spacecraft!

Review Questions

Q1. Name the four main parts of a desktop computer setup.

Q2. What is the difference between hardware and software? Give two examples of each.

Q3. Classify these devices as Input or Output: Printer, Keyboard, Scanner, Speakers, Mouse.

Q4. What does CPU stand for, and what is its role?

Q5. Can a computer work with only hardware and no software? Explain why or why not.

Challenge Task

Draw a labelled diagram of a computer setup on a blank sheet of paper. Include at least **six** parts and mark each one as either "Input," "Output," or "Processing." Add colour to make it look like a poster you'd hang in a computer lab!

3 Types of Computers

Learning Objectives

- Identify the four main types of personal computers
- Compare desktop, laptop, tablet, and smartphone features
- Understand which type of computer is best for different tasks
- Recognise that all these devices are computers

Did you know? The device in your pocket — yes, your smartphone — is a computer too! It just happens to fit in your jeans. Cool, right?

3.1 Different Types of Computers

Not all computers look the same. They come in many shapes and sizes, each designed for different purposes. Let's explore the four types you'll encounter most often:

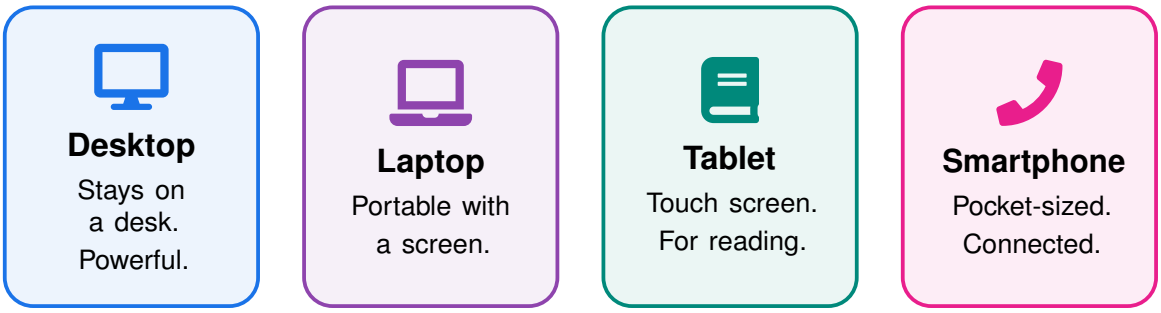


Figure 3.1: The four main types of personal computers

3.2 Comparing Computer Types

Each type of computer has its own strengths and weaknesses. Let's compare them:

Table 3.1: Comparing different types of computers

Feature	Desktop	Laptop	Tablet	Smartphone
Portability	Low	Medium	High	Very High
Screen Size	Large	Medium	Medium	Small
Performance	High	Medium	Medium	Low–Medium
Battery	✗ No	✓ Yes	✓ Yes	✓ Yes
Best For	School lab, office	Travel, homework	Reading, browsing	Calls, messaging

Note

All four types — desktop, laptop, tablet, and smartphone — are **computers**. They all accept input, process data, store information, and produce output. They just do it in different shapes and sizes!

3.3 Which Computer Should You Use?

It depends on what you need to do! Here's a quick guide:

- **Writing a long essay?** Desktop or laptop — a real keyboard makes typing faster.
- **Reading an e-book on the bus?** Tablet — light, easy to hold, great screen.
- **Calling a friend?** Smartphone — it's designed for communication.
- **Playing a high-end game?** Desktop — most powerful option for gaming.

Try It Yourself!

Look around your classroom or home. Make a list of every computer-like device you can find. Don't forget things like smart TVs, gaming consoles, and smartwatches — they're all computers too!

Smartphones are computers too! They just fit in your pocket. The phone you carry has more computing power than the computers used for the entire Apollo 11 Moon landing mission!

Did You Know?

The first laptop was the **Osborne 1**, released in 1981. It weighed about **11 kilograms** (that's heavier than a bowling ball!) and had a tiny 5-inch screen. Today's laptops weigh less than 1.5 kg and have screens up to 17 inches. That's a LOT of progress!

Review Questions

Q1. Name four types of personal computers.

Q2. Which type of computer is *most portable*? Which is *least portable*?

Q3. Why would someone choose a desktop computer over a laptop?

Q4. Is a smartphone a computer? Explain your answer.

Q5. Give one example of a task best suited for each type of computer.

Challenge Task

Create a **poster** comparing the four types of computers. For each one, draw a simple picture, list 2 advantages and 1 disadvantage, and write one sentence about when you'd use it. Make it colourful!

4 Basics of Software

Learning Objectives

- Define software and explain why computers need it
- Distinguish between system software and application software
- Give examples of common system and application software
- Understand how hardware and software work together

If hardware is the computer's **body**, then software is its **mind**. Without software, a computer is just an expensive box of metal and plastic!

4.1 What Is Software?

Software is a set of instructions (programs) that tells the computer hardware what to do. You can't touch software — it exists as code stored in the computer's memory.

Definition: Software

Software is a collection of programs, data, and instructions that tell the computer how to perform specific tasks. Unlike hardware, software has no physical form — you can't pick it up or hold it.

4.2 Types of Software

Software is divided into two main categories:

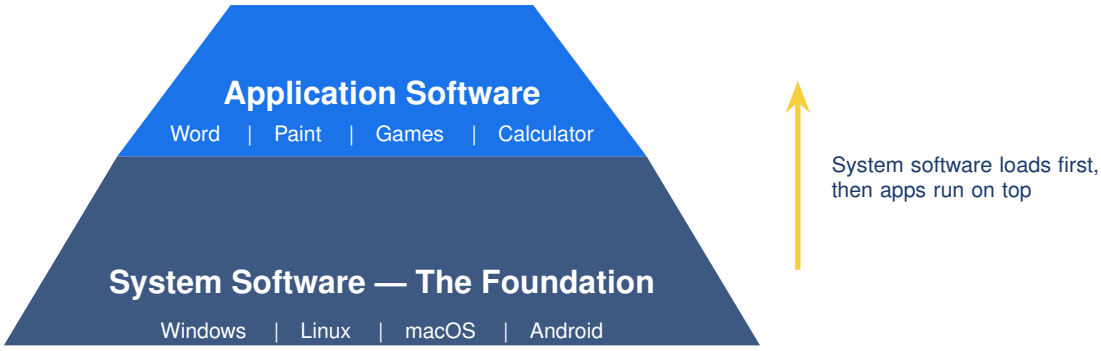


Figure 4.1: Software types: system software supports application software

4.2.1 System Software

System software manages the computer's hardware and provides a platform for other software to run. The most important system software is the **Operating System** (OS).

Examples: **Windows**, **macOS**, **Linux**, **Android**, **iOS**

4.2.2 Application Software

Application software (or "apps") are programs you use to perform specific tasks — like writing a document, browsing the internet, or playing a game.

Examples: **MS Word**, **MS Paint**, **Calculator**, **Google Chrome**, **Minecraft**

4.3 Hardware + Software = A Working Computer!

Neither hardware nor software can work alone. They need each other!

Figure 4.2: Hardware and software must work together

Real World Connection

Think of it like a music player. The **hardware** is the speaker and screen. The **software** is the music app. Without the app, the speaker stays silent. Without the speaker, the app can't play any sound. They *need* each other!

4.4 Common Software Examples

Table 4.1: Common software examples and what they do

Type	Name	What It Does
System Software	Windows 11	Manages the computer
System Software	Android	Runs your smartphone
Application	MS Word	Word processing (writing documents)
Application	MS Paint	Drawing & image editing
Application	Calculator	Maths operations
Application	Chrome	Browse the internet

Every time you open an app on your phone, you're running **application software** on top of **system software** (Android or iOS). It's like a building: the OS is the foundation, and apps are the rooms you actually live in!

Did You Know?

The first ever software bug was an *actual* bug! In 1947, engineers found a moth stuck inside a computer at Harvard University. They taped it into their logbook and wrote "first actual case of bug being found." That's how the term "computer bug" was born!

Review Questions

Q1. What is software? How is it different from hardware?

Q2. Name the two main types of software.

Q3. Give three examples of **system software**.

Q4. Give four examples of **application software**.

Q5. Can hardware work without software? Explain your answer.

Q6. What is an operating system, and why is it important?

Challenge Task

Make a two-column table in your notebook. In the left column, list **10 software programs** you've used. In the right column, classify each one as either "System Software" or "Application Software." Bonus: add a third column describing what each program does!

5 Computer Lab Rules & Safety

Learning Objectives

- List the important rules for using a computer lab safely
- Demonstrate correct sitting posture at a computer desk
- Explain why lab safety is important for health and equipment
- Identify common mistakes students make in a computer lab

Your computer lab is a special place — treat it with care, and it (and your body!) will thank you. Let's learn how to be a safe and responsible computer user!

5.1 Why Do We Need Lab Rules?

A computer lab is full of expensive equipment that can be easily damaged. The rules exist to:

- Keep **you** safe from electrical hazards and health problems
- Keep the **computers** safe from food spills, rough handling, and viruses
- Make the lab a **pleasant place** for everyone to work

5.2 Correct Sitting Posture

Sitting incorrectly at a computer can cause back pain, neck strain, and tired eyes. Here's how to sit properly:

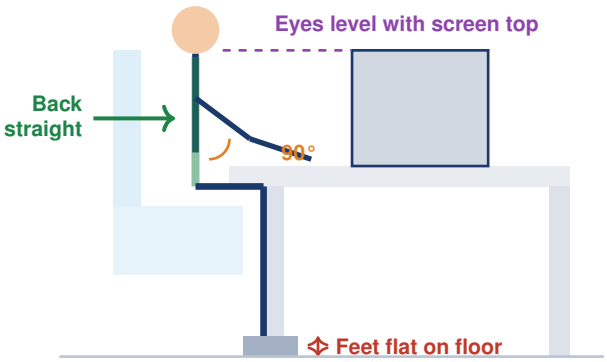


Figure 5.1: Correct sitting posture at a computer desk

Warning!

Bad posture habits can cause:

- Back and neck pain
- Headaches from eye strain
- Wrist pain (from typing at wrong angles)
- Tiredness and difficulty concentrating

Always sit up straight and take a short break every 30 minutes!

5.3 Lab Rules Everyone Must Follow

Figure 5.2: Six essential computer lab rules

Warning!

Never force-pull cables or plug in USB drives the wrong way around — always check the connector shape first! Forcing a cable can damage both the cable and the port permanently.

BYTE

If something goes wrong — a strange error, a weird noise, or a screen that won't turn on — **don't panic!** Tell your teacher immediately. Trying to fix it yourself could make things worse.

Did You Know?

Some schools in Japan have "computer lab shoes" — students change into clean indoor slippers before entering the lab to keep dust and dirt away from the equipment. It might sound extreme, but dust is one of the biggest enemies of computers!

Review Questions

Q1. List any five rules you should follow in a computer lab.

Q2. Why should you not eat or drink near a computer?

Q3. Describe the correct sitting posture for using a computer.

Q4. What should you do if a computer shows an error or stops working?

Q5. Why is it important to shut down a computer properly instead of just switching it off?

Challenge Task

Design a **lab rules poster** for your computer lab. Include at least 6 rules with a simple icon or drawing for each one. Use bright colours and make it eye-catching enough that visitors would stop and read it!

6 Part I — Hands-On Practicals

BYTE

Time to get your hands on a real computer! Remember — read each step carefully, ask your teacher if you're stuck, and most importantly, have fun!

Practical 1: Identifying Computer Parts 10 min

You Will Need: Computer or printed image | 10 minutes | Worksheet

Step Map:

Objective: Identify the main parts of a computer by name and function.

Materials: Real computer OR printed A3 image of a computer setup; identification worksheet.

Steps:

1. Look at the computer (or image) on your desk.
2. Point to and name each part your teacher asks about.
3. Fill in the *Parts Identification Worksheet* — draw a line from each label to the correct part.
4. Answer: Is a keyboard an input or output device?
5. Answer: What does the monitor display?

Expected Outcome: Students correctly label Monitor, CPU, Keyboard, Mouse on the worksheet.

Practical 2: Starting and Shutting Down a Computer 10 min

You Will Need: Working computer 10 minutes Observation sheet

Step Map:

Objective: Learn how to safely start and shut down a computer.

Steps:

1. Press the power button on the CPU tower. Notice the light turns on.
2. Wait patiently for the desktop to appear. This may take 30–60 seconds.
3. Look at the desktop. Can you see icons? The taskbar? The Start button?
4. Click the **Start** button → Click **Power** → Click **Shut Down**.
5. Wait until the screen goes completely dark and the power light turns off.

Expected Outcome: Students can independently start and shut down a computer safely.

Practical 3: Mouse Practice — Click, Double-Click, Drag 15 min

You Will Need: Computer with mouse | 15 minutes | Mouse

Step Map:

Objective: Master single click, double-click, right-click, and drag-and-drop.

Steps:

1. **Single Click:** Click once on the Recycle Bin icon. It should become highlighted (selected).
2. **Double-Click:** Double-click the Recycle Bin to open it. A window should appear.
3. Close the window by clicking the **X** button in the top-right corner.
4. **Right-Click:** Right-click on an empty part of the desktop. A menu should appear.
5. Press `Esc` to close the menu.
6. **Drag and Drop:** Click and hold any desktop icon, drag it to a new position, then release.

Expected Outcome: Students can perform all four mouse actions confidently.

Practical 4: Keyboard Exploration 10 min

You Will Need: Computer 10 minutes Keyboard

Objective: Identify key groups on the keyboard and type your name.

Steps:

1. Open **Notepad** (Start → search "Notepad" → click to open).
2. Type your full name. Press `Enter` to go to a new line.
3. Type the name of your school on the new line.
4. Find and press the `Caps Lock` key. Type your name again — notice it's ALL CAPITALS.
5. Press `Caps Lock` again to turn it off.
6. Find the `Backspace` key and use it to delete some text.
7. Close Notepad *without* saving (click "Don't Save" when asked).

Expected Outcome: Students can type text, use Caps Lock, Enter, and Backspace.

Practical 5: Exploring the Desktop 10 min

You Will Need: Computer 10 minutes Observation sheet

Objective: Identify elements of the Windows desktop.

Steps:

1. Look at your desktop. Find and write down at least 3 icon names.
2. Find the **Taskbar** at the bottom. What icons do you see pinned there?
3. Click the **Start button** (Windows icon, bottom-left). What appears?
4. Look at the bottom-right corner. Find the **clock**, **volume** icon, and **Wi-Fi** icon.
5. Right-click on an empty part of the desktop. Explore the options in the menu that appears.

Expected Outcome: Students can identify and describe desktop icons, taskbar, Start menu, and system tray.

Practical 6: Classifying Hardware and Software 10 min

You Will Need: Computer 10 minutes Classification worksheet

Objective: Create a list and classify items as hardware or software.

Steps:

1. Look around the computer lab. Write down 5 things you can **physically touch** (these are **hardware**).
2. Now think of 5 programs or apps (these are **software**).
3. On your worksheet, draw a table with two columns: “Hardware” and “Software.”
4. Place each item in the correct column.
5. Share your lists with a partner and check each other’s work.

Expected Outcome: Students correctly classify at least 8 out of 10 items.

Scan for extra Part I resources and activities!

PART II

Windows Operating System

Learn to navigate Windows like a pro — from the desktop to file management and beyond.

Windows Files Settings Security Troubleshooting

Part II

Windows Operating System

7 Introduction to Operating Syster

Learning Objectives

- Define what an operating system is and why it is needed
- Explain the key functions of an operating system
- Name popular operating systems used today
- Understand the relationship between hardware, OS, and applications

An operating system is like the **manager** of a restaurant. You (the customer) don't talk to the chef (hardware) directly — the manager (OS) takes your order and makes sure everything runs smoothly!

7.1 What Is an Operating System?

An operating system (OS) is the most important software on a computer. It manages all the hardware and software, and provides the interface you see and interact with (like the desktop, icons, and windows).

Definition: Operating System

An **Operating System (OS)** is system software that:

- Manages the computer's **hardware** (CPU, memory, storage, peripherals)
- Provides a **user interface** (desktop, menus, windows)
- Runs and manages **application software**
- Handles **files and folders** on your storage drives
- Provides **security** (user accounts, passwords, firewall)

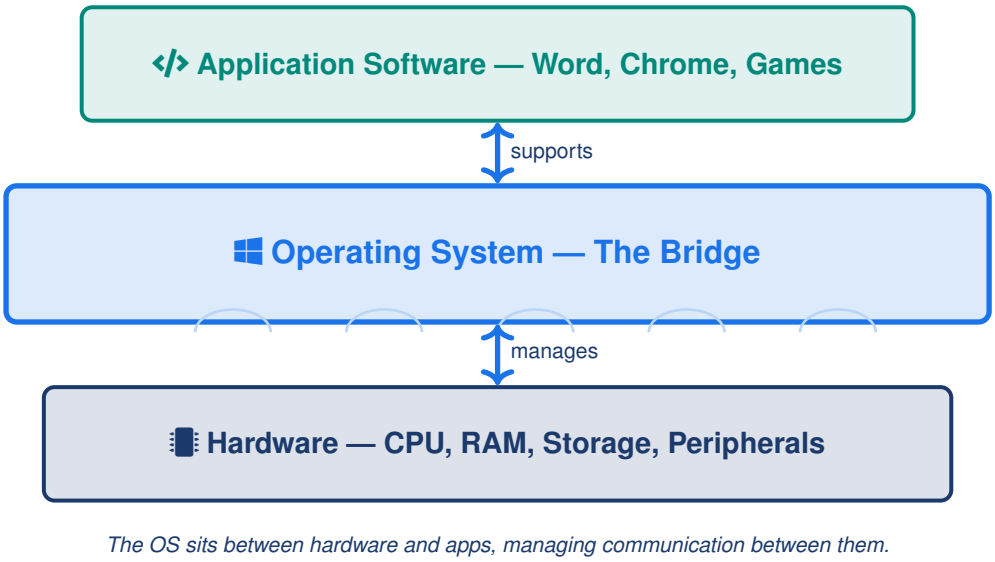


Figure 7.1: The operating system acts as a bridge between hardware and applications

7.2 A Brief History of Windows

Microsoft Windows is the most widely used operating system in the world. Here's how it evolved:

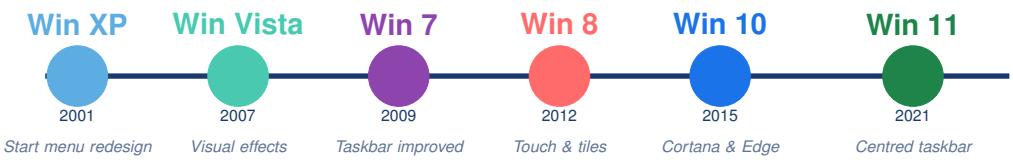


Figure 7.2: A timeline of major Windows versions

7.3 Functions of an Operating System

Table 7.1: Key functions of an operating system

	Function	What the OS Does	Example
	Memory	Manages RAM allocation	Opens Chrome
	Storage	Reads/writes files	Saves a document
	Interface	Shows you the desktop	Icons, Start Menu
	Network	Connects to the Internet	Wi-Fi management
	Security	Protects against threats	Windows Defender

BYTE

There are many operating systems besides Windows! **macOS** runs on Apple MacBooks, **Linux** is free and open-source (used by hackers and scientists), and **Android** runs on most smartphones. They all do the same basic job but look and feel different.

Did You Know?

The name “Windows” comes from the fact that the operating system displays information in rectangular areas on screen called **windows**. Before Windows, people used MS-DOS, where you had to type commands in plain text — no mouse, no icons, no colours! Imagine using a computer with only a blank screen and a blinking cursor. Scary, right?

Review Questions

Q1. What is an operating system? Write a definition in your own words.

Q2. Name three popular operating systems.

Q3. List four functions of an operating system.

Q4. What was the operating system used before Windows?

Q5. Why is the OS called a “bridge” between hardware and software?

Challenge Task

Research the history of **one operating system** (Windows, macOS, Linux, or Android). Write a short paragraph about when it was first released, who created it, and one interesting fact about it. Present your findings to the class!

8 Starting and Shutting Down Win

Learning Objectives

- Describe the boot process when a computer starts
- Explain the difference between Shut Down, Restart, and Sleep
- Safely start and shut down a Windows computer
- Understand what happens during the POST check

Starting a computer isn't as simple as pressing a button — there's a whole sequence of events happening behind the scenes! Let's peek under the hood.

8.1 The Boot Process

When you press the power button, the computer goes through a series of steps before the desktop appears. This is called the boot process.

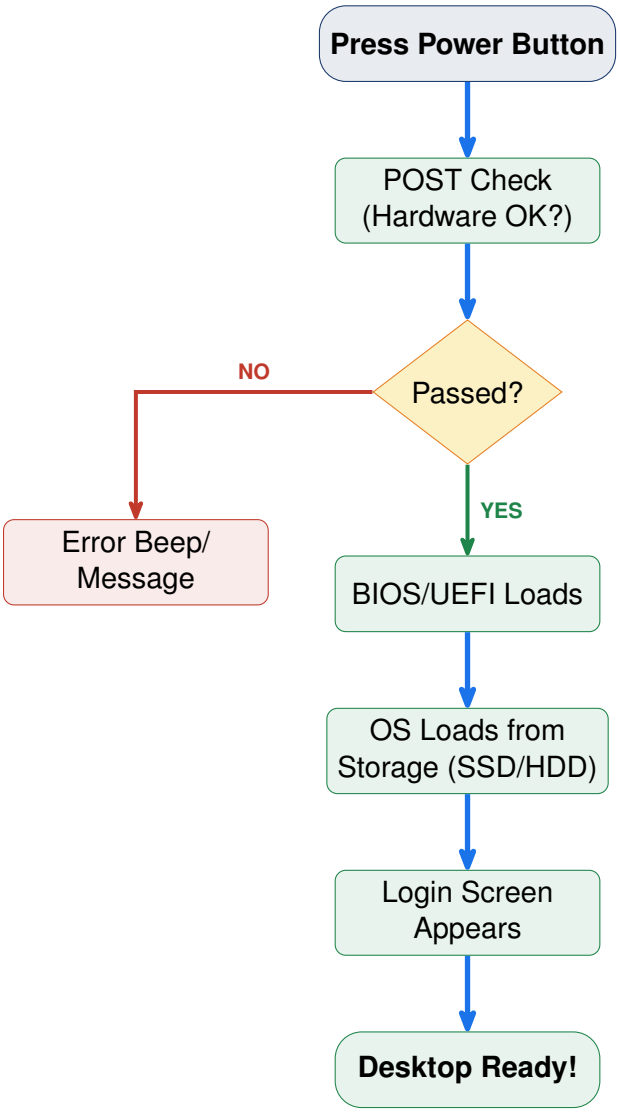


Figure 8.1: The boot process — from power button to desktop

POST — Power-On Self-Test

POST stands for Power-On Self-Test. It's a quick check the computer performs when it starts up to make sure all the essential hardware (RAM, keyboard, storage) is working. If something is wrong, you'll hear beeping sounds or see an error message.

8.2 Shutdown Options

When you're done using the computer, you have three choices:

Shut Down

Turns off completely. All programs close. Use at end of day.

Restart

Turns off, then turns back on. Use after updates.

Sleep

Low power mode. Work is saved in RAM. Use for short breaks.

Figure 8.2: Three ways to power off — choose the right one!

Warning!

Never turn off a computer by holding the power button (force shutdown) unless it's completely frozen and won't respond. Force shutting down can cause data loss or even damage files!

BYTE

Always **save your work** before shutting down or restarting! Unsaved files will be lost. Get into the habit of pressing `Ctrl`+`S` regularly.

Did You Know?

Early computers in the 1960s had no "shut down" option. Operators literally had to flip large physical switches to turn them off, and the boot-up process could take **hours**! Today, your computer can start in under 10 seconds with an SSD.

Review Questions

Q1. What does POST stand for, and what does it do?

Q2. List the steps of the boot process in order.

Q3. What is the difference between Shut Down and Sleep?

Q4. When should you use "Restart" instead of "Shut Down"?

Q5. Why should you never force-shutdown a computer unless it's frozen?

Challenge Task

Next time you turn on a computer, **time the boot process** with a stopwatch. How long does it take from pressing the power button to seeing the desktop? Compare with a classmate. Who has the faster computer, and why do you think that is?

9 Windows Desktop Environment

Learning Objectives

- Identify the main elements of the Windows desktop
- Explain the purpose of the Taskbar, Start Menu, and System Tray
- Use desktop icons and shortcuts effectively
- Navigate the Windows desktop with confidence

The desktop is your computer's "home screen" — like the home screen on a phone, but with WAY more power. Let's explore every corner of it!

9.1 The Windows Desktop

When Windows finishes loading, you see the desktop. This is your main workspace — everything starts here.

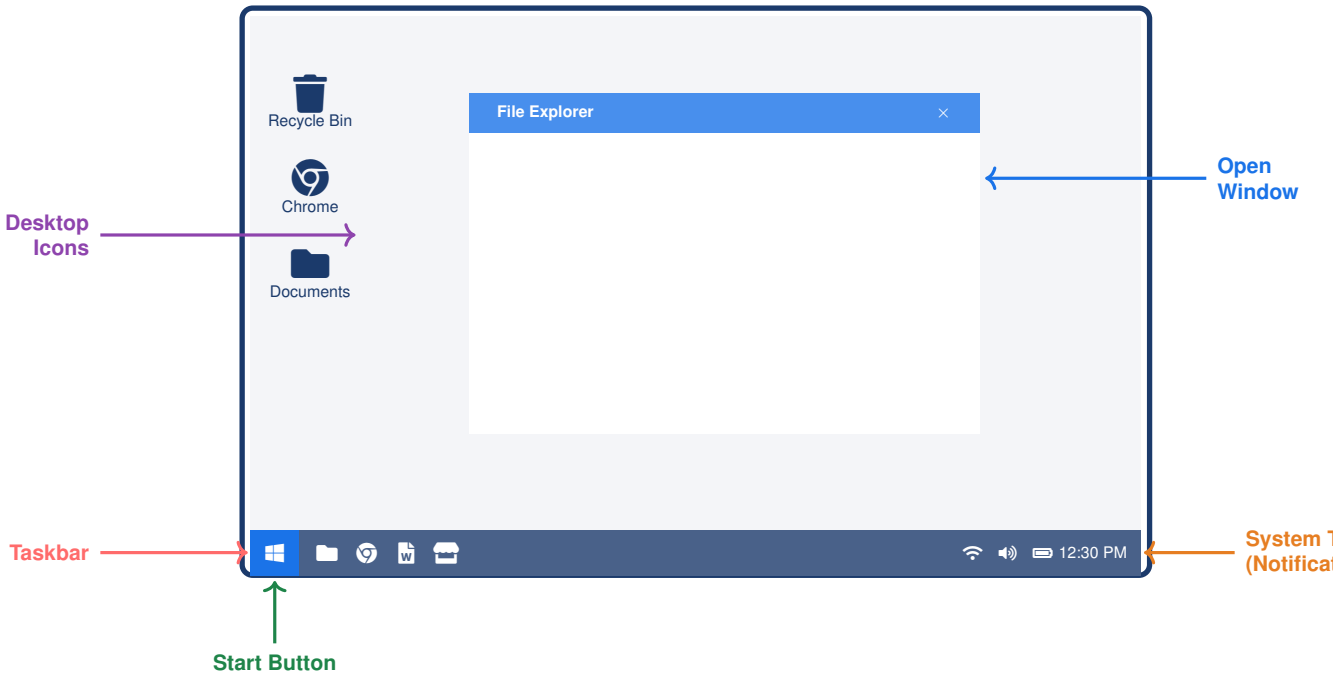


Figure 9.1: The Windows desktop with all main elements labelled

Table 9.1: Desktop elements quick reference

Icon	Element	What It Does
	Desktop	Main working area where icons and windows appear
	Taskbar	Shows open programs and pinned apps at the bottom
	Start Menu	Opens apps, settings, power options, and search
	Icons	Shortcuts to files, folders, and programs
	Notification Area	Alerts, date/time, Wi-Fi, volume, and battery
	Recycle Bin	Deleted files go here (can be recovered!)

Note

The **Recycle Bin** is like a rubbish bin for your computer. When you delete a file, it goes to the Recycle Bin first. You can restore it if you change your mind! Files are only permanently deleted when you **empty** the Recycle Bin.

BYTE

You can customise your desktop! Right-click on an empty area of the desktop and try "Personalise" to change the wallpaper, colours, and theme. Make your computer feel like *yours*!

Did You Know?

The Recycle Bin icon on your desktop actually changes its appearance! When it's empty, it shows an empty bin. When it contains deleted files, it shows a bin with crumpled paper inside. Nice detail, right?

Review Questions

Q1. Name five elements of the Windows desktop.

Q2. What is the purpose of the Taskbar?

Q3. How do you access the Start Menu?

Q4. What is the Recycle Bin, and where do deleted files go?

Q5. What information can you find in the System Tray (notification area)?

Challenge Task

Customise your desktop: change the wallpaper, rearrange the icons, and pin your favourite app to the taskbar. Then take a screenshot (`PrtSc` key) and paste it into MS Paint. Save it as "MyDesktop.png."

10 File and Folder Management

Learning Objectives

- Understand the concept of files and folders
- Create, rename, copy, move, and delete files and folders
- Navigate the File Explorer window
- Know the rules for naming files in Windows

Think of your computer like a giant filing cabinet. **Folders** are the drawers, and **files** are the documents inside them. Keeping things organised will save you *hours* of searching!

10.1 What Are Files and Folders?

Files and Folders

- A **file** is a collection of data stored with a name and extension (e.g., `MyEssay.docx`, `Photo.jpg`)
- A **folder** (also called a **directory**) is a container that holds files and other folders
- Folders help you **organise** your files so you can find them easily

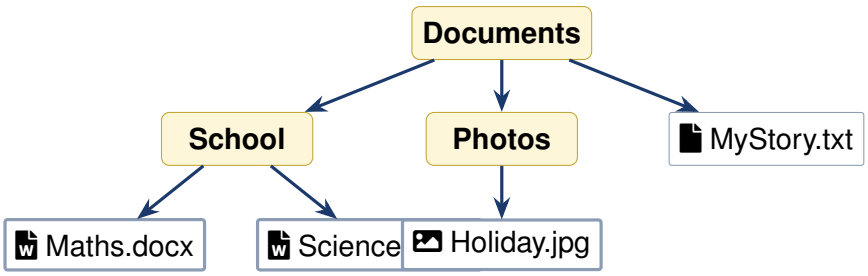


Figure 10.1: Files and folders form a tree-like hierarchy

10.2 File Operations

Here are the common things you can do with files:

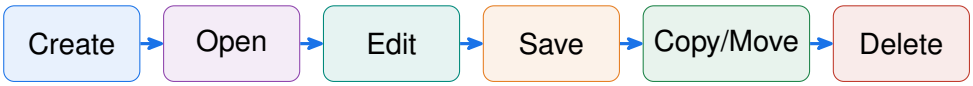


Figure 10.2: Common file operations in sequence

Keyboard Shortcuts

Ctrl + C	Copy selected file
Ctrl + X	Cut (move) selected file
Ctrl + V	Paste file
Delete	Send file to Recycle Bin
F2	Rename selected file
Ctrl + Z	Undo last action

10.3 File Naming Rules

Table 10.1: Valid vs invalid file names in Windows

✓ Valid Names	✗ Invalid Names
MyReport	My/Report (slash not allowed)
Science_Notes_2024	File:Name (colon not allowed)
Chapter 1 Draft	CON (reserved name in Windows)
photo_holiday	file* (asterisk not allowed)

Never delete files from the **Windows** or **System32** folder. These contain critical files that your computer needs to work. Deleting them can *break your entire computer*!

Note

Windows file names **cannot contain** these characters: `\ / : * ? " < > |`. Also, names like CON, PRN, AUX, NUL are reserved by Windows and can't be used.

Did You Know?

The longest file name allowed in Windows is **255 characters** (including the extension). But the *full path* (including all folder names) can't exceed about 260 characters. So if you nest folders very deeply with long names, you might run into problems!

Review Questions

Q1. What is the difference between a file and a folder?

Q2. List three operations you can perform on a file.

Q3. Name three characters that cannot be used in a Windows file name.

Q4. What is the keyboard shortcut for copying a file?

Q5. What happens when you press the `Delete` key on a selected file?

Challenge Task

Create the following folder structure on a computer: `MyProject > Documents`, `MyProject > Images`, `MyProject > Videos`. Inside Documents, create a text file called "Notes.txt." Then copy it to the Images folder and rename the copy to "ImageNotes.txt."

11 Using Windows Applications

Learning Objectives

- Open and use built-in Windows applications
- Identify the purpose of Notepad, WordPad, Paint, and Calculator
- Navigate a basic application window (title bar, menus, toolbar)
- Save and close work in an application

BYTE

Windows comes with free apps already installed — like getting free games with a new phone! Let's explore the cool tools you already have.

11.1 Built-In Windows Applications

Windows comes with several useful applications pre-installed. Here are four you'll use often:

Notepad
Simple text editor for quick notes.
Best for: Plain text

WordPad
Basic word processor with formatting.
Best for: Short docs

Paint
Drawing & basic image editing.
Best for: Drawings

Calculator
Maths operations and conversions.
Best for: Maths

Figure 11.1: Four built-in Windows applications you should know

11.2 Understanding Application Windows

Every application opens in a **window**. All windows share common elements:

Parts of an Application Window

- **Title Bar** — Shows the file name and app name; has Minimise, Maximise, and Close buttons
- **Menu Bar** — Contains menus like File, Edit, Format, View, Help

- **Toolbar / Ribbon** — Quick-access buttons for common tasks
- **Work Area** — The main area where you type, draw, or view content
- **Status Bar** — Shows helpful info at the bottom (zoom, line count, etc.)

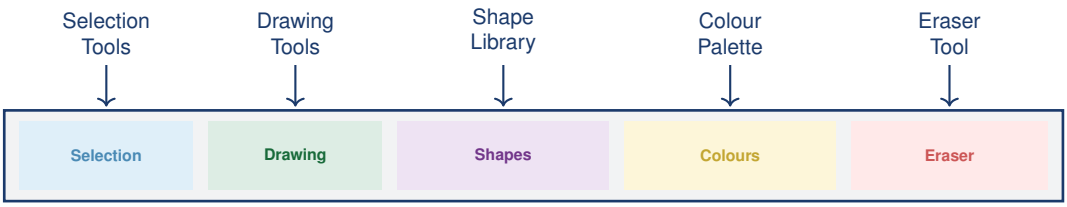


Figure 11.2: The Paint toolbar — key sections labelled

To open any app quickly, press the `Windows` key and start typing the app's name. Windows Search will find it instantly — no need to click through menus!

Try It Yourself!

Open **MS Paint** and draw a simple house with a door, two windows, and a sun in the sky. Use at least **three different colours**. Save your masterpiece as "MyHouse.png" on the Desktop!

Did You Know?

MS Paint has been included with every version of Windows since **Windows 1.0 in 1985**. That makes it over 40 years old! Microsoft once threatened to remove it, but there was such a public outcry that they kept it. People *love* Paint!

Review Questions

Q1. Name four built-in Windows applications and describe what each does.

Q2. What are the main parts of an application window?

Q3. How do you open an application using Windows Search?

Q4. What is the difference between Notepad and WordPad?

Q5. How do you close an application window?

Challenge Task

Open the **Calculator** app and use it to solve these problems: (a) 256×48, (b) $7894 \div 23$, (c) 15% of 680. Write down your answers. Then switch to **Scientific**

mode (click the menu $\rightarrow$ Scientific) and calculate 2^{10}. What do you get?

12 Control Panel and Settings

Learning Objectives

- Access the Windows Settings and Control Panel
- Change basic system settings (display, date/time, volume)
- Understand the purpose of common Control Panel categories
- Personalise the computer's appearance

The Control Panel and Settings are like the "command centre" of your computer. This is where you tweak, tune, and customise everything!

12.1 Settings vs Control Panel

Windows has two places to change settings:

- **Settings** (modern, touch-friendly) — press `Windows`+`I`
- **Control Panel** (classic, more detailed) — search "Control Panel" in Start

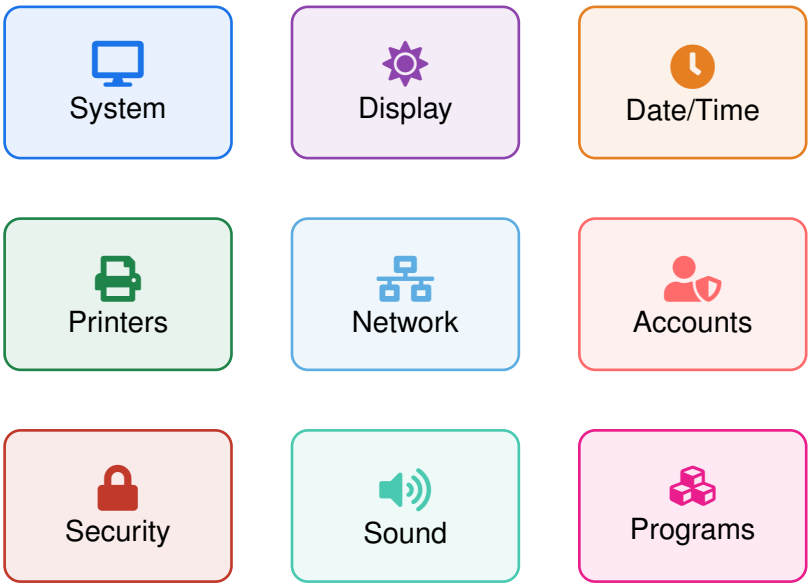


Figure 12.1: Control Panel categories — your computer's command centre

12.2 Common Settings You Can Change

1. **Display Settings** — Change screen brightness, resolution, and text size
2. **Date and Time** — Set the correct date, time, and time zone
3. **Sound Settings** — Adjust volume, choose notification sounds
4. **Personalisation** — Change wallpaper, colours, lock screen image
5. **User Accounts** — Create accounts, set passwords, manage users

Be careful when changing settings you don't understand! Some changes (like network or security settings) can stop your internet from working or lock you out. If in doubt, ask your teacher.

Did You Know?

The Control Panel has been part of Windows since **Windows 3.0** in 1990. Microsoft has been slowly moving features into the newer "Settings" app, but the classic Control Panel still exists and is used by IT professionals every day!

Review Questions

Q1. What is the keyboard shortcut to open Windows Settings?

Q2. Name four categories found in the Control Panel.

Q3. How would you change your desktop wallpaper?

Q4. What is the difference between Settings and Control Panel?

Q5. Why should you be careful when changing system settings?

Challenge Task

Open **Settings** → **Personalisation** and change your desktop wallpaper to a solid colour. Then change the system accent colour to match. Finally, change it all back to what it was before!

13 Storage Devices

Learning Objectives

- Name and describe different types of storage devices
- Compare storage devices by capacity, speed, and portability
- Understand the difference between primary and secondary storage
- Explain what cloud storage is and how it works

Storage is where your computer keeps everything — from your homework to your favourite photos. Think of it like a digital wardrobe for all your stuff!

13.1 Types of Storage

Primary vs Secondary Storage

- **Primary storage** (RAM) — Temporary, fast, loses data when powered off
- **Secondary storage** (HDD, SSD, USB) — Permanent, keeps data even when off

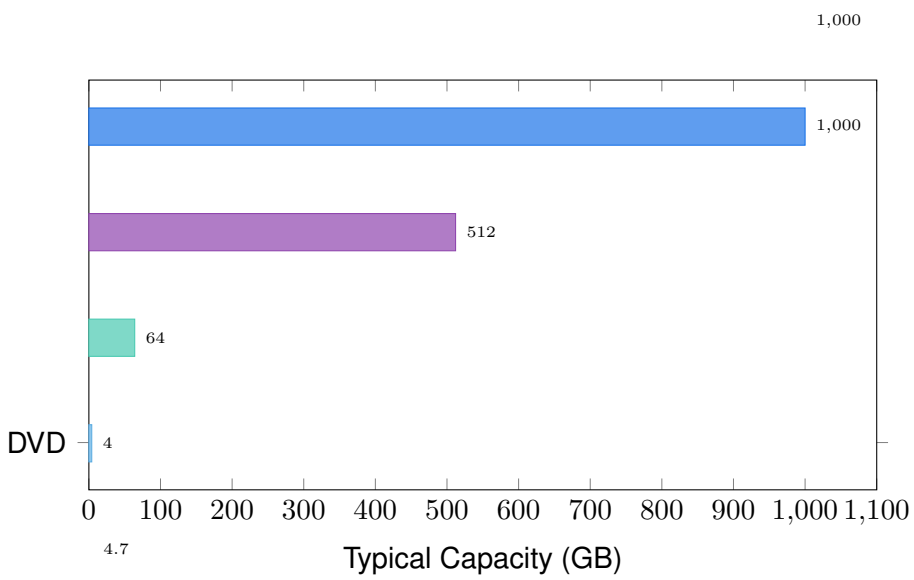


Figure 13.1: Comparing storage device capacities (Cloud shown as 1000+ GB)

Table 13.1: Comparing different storage devices

Device	Type	Capacity	Speed	Portable
HDD	Magnetic	500 GB – 4 TB	Medium	✗
SSD	Flash	128 GB – 2 TB	Fast	✗
USB Drive	Flash	4 GB – 256 GB	Medium	✓
DVD	Optical	4.7 GB	Slow	✓
Cloud	Online	Unlimited	Varies	✓

13.2 Cloud Storage

Cloud storage lets you save files on the internet instead of your computer. Services like **Google Drive**, **OneDrive**, and **Dropbox** let you access your files from any device, anywhere!

Real World Connection

When you save a photo to Google Photos or upload a document to Google Drive, you're using cloud storage. The file isn't on your phone or laptop — it's on a powerful computer (called a **server**) in a data centre, possibly thousands of kilometres away!

If you stacked all the data stored in the world's cloud servers as printed paper, the pile would reach from Earth to Pluto — and back — **multiple times**!

Did You Know?

The first USB flash drive was sold in 2000 and could store just **8 MB** of data — that's not even enough for one high-quality photo today! Modern USB drives can hold up to **2 TB**, which is 250,000 times more.

Review Questions

Q1. What is the difference between primary and secondary storage?

Q2. Name four types of secondary storage devices.

Q3. Which storage device is the fastest: HDD, SSD, or DVD?

Q4. What is cloud storage, and name two cloud storage services.

Q5. Why is an SSD faster than an HDD?

Challenge Task

Check how much free storage your school computer has. Right-click on **This PC** → **Properties** (or check the C: drive). Write down: (a) Total capacity, (b) Used space, (c) Free space. Calculate what percentage is used!

14 Connecting to the Internet

Learning Objectives

- Understand what an internet connection is
- Connect a Windows computer to a Wi-Fi network
- Check the status of your internet connection
- Troubleshoot basic connection problems

The internet connects your computer to the whole world. Let's learn how to get online — and what to do if the connection drops!

14.1 Types of Internet Connections

Internet Connection Methods

- **Wi-Fi** — Wireless connection via a router (most common at home and school)
- **Ethernet** — Wired connection using a cable (faster and more reliable)
- **Mobile Data** — Uses your phone's cellular network (3G/4G/5G)
- **Broadband** — High-speed connection via phone line or fibre optic

14.2 Connecting to Wi-Fi

To connect to Wi-Fi on Windows:

1. Click the **Wi-Fi icon** in the System Tray (bottom-right).
2. A list of available networks will appear.
3. Click on your school or home network name.
4. Enter the **password** if required and click Connect.
5. A "Connected" message confirms you're online!

Note

If you see a 🔒 padlock icon next to a network name, it means the network is **password-protected**. You need the correct password to join. Open networks (without a padlock) don't require a password, but they are less secure.

Be careful with **public Wi-Fi** (in cafes, airports, malls). These networks are not secure — someone could spy on your activity. Never log into your bank account or enter passwords on public Wi-Fi!

Did You Know?

The word "Wi-Fi" doesn't actually stand for anything! Many people think it means "Wireless Fidelity," but the Wi-Fi Alliance has confirmed that it's just a catchy brand name they invented. It sounds tech-y, and that was enough!

Review Questions

Q1. Name three types of internet connections.

Q2. What steps do you follow to connect to Wi-Fi on Windows?

Q3. What does a padlock icon next to a Wi-Fi network mean?

Q4. Why should you be careful with public Wi-Fi?

Q5. What is the difference between Wi-Fi and Ethernet?

Challenge Task

Check the Wi-Fi connection on your school computer. What is the network name? Is it password-protected? Open a web browser and go to `speedtest.net` to test your internet speed. Write down the **download speed** and **upload speed**.

15 Computer Security Basics

Learning Objectives

- Understand why computer security is important
- Create strong passwords and recognise weak ones
- Identify common threats: viruses, malware, phishing
- Use Windows security features to stay safe

Your computer is like your house — you wouldn't leave the front door wide open, right? Computer security is about **locking the digital doors**!

15.1 Why Security Matters

Every day, millions of computers are targeted by hackers, viruses, and scams. Good security habits protect your personal information, school work, and the computer itself.

15.2 Password Strength

Not all passwords are equal. Here's how to judge password strength:

Figure 15.1: Password strength meter — aim for the green zone!

Creating a Strong Password

A strong password should have:

- ✓ At least **8 characters** (12+ is better)
- ✓ A mix of **uppercase** and **lowercase** letters
- ✓ At least **one number** (0–9)

✓ At least **one special character** (, @, #, $, %, etc.)

✓ **No personal info** (not your name, birthday, or "password")

15.3 Common Threats

⚠ **Virus** — A program that spreads by infecting other files and can damage your system

⚠ **Malware** — Any malicious software (viruses, spyware, ransomware)

⚠ **Phishing** — Fake emails or websites that trick you into sharing passwords

⚠ **Spam** — Unwanted junk messages, often containing scams or malware

⚠ Warning!

Never click links in emails from unknown senders! Phishing emails often look like they're from real companies (banks, schools, social media). Always check the sender's email address carefully.

Windows has a built-in security tool called **Windows Defender**. Make sure it's always turned on! It scans for viruses and blocks threats automatically. You can check it in Settings → Privacy & Security → Windows Security.

★ Did You Know?

The first computer virus was called **"Creeper"** and was created in 1971 — just as an experiment! It displayed the message: "I'm the creeper, catch me if you can!" A program called "Reaper" was written to delete it. That makes Reaper the world's first antivirus!

? Review Questions

Q1. What makes a strong password? List four characteristics.

Q2. What is a computer virus?

Q3. How is phishing different from a virus?

Q4. Name the built-in security tool in Windows.

Q5. Give three tips for staying safe online.

Challenge Task

Create **three strong passwords** using the rules you learned. Each must be at least 12 characters and include uppercase, lowercase, numbers, and symbols. Write them down (but don't use them for real accounts — this is just practice!).

16 Basic Troubleshooting

Learning Objectives

- Identify common computer problems
- Follow a logical troubleshooting process
- Fix simple issues like frozen screens and no sound
- Know when to ask for help from a teacher or IT support

Things go wrong sometimes — that's perfectly normal! The important thing is knowing how to fix them calmly and logically. Let's learn the troubleshooting mindset!

16.1 The Troubleshooting Flowchart

When something goes wrong, follow this step-by-step process:

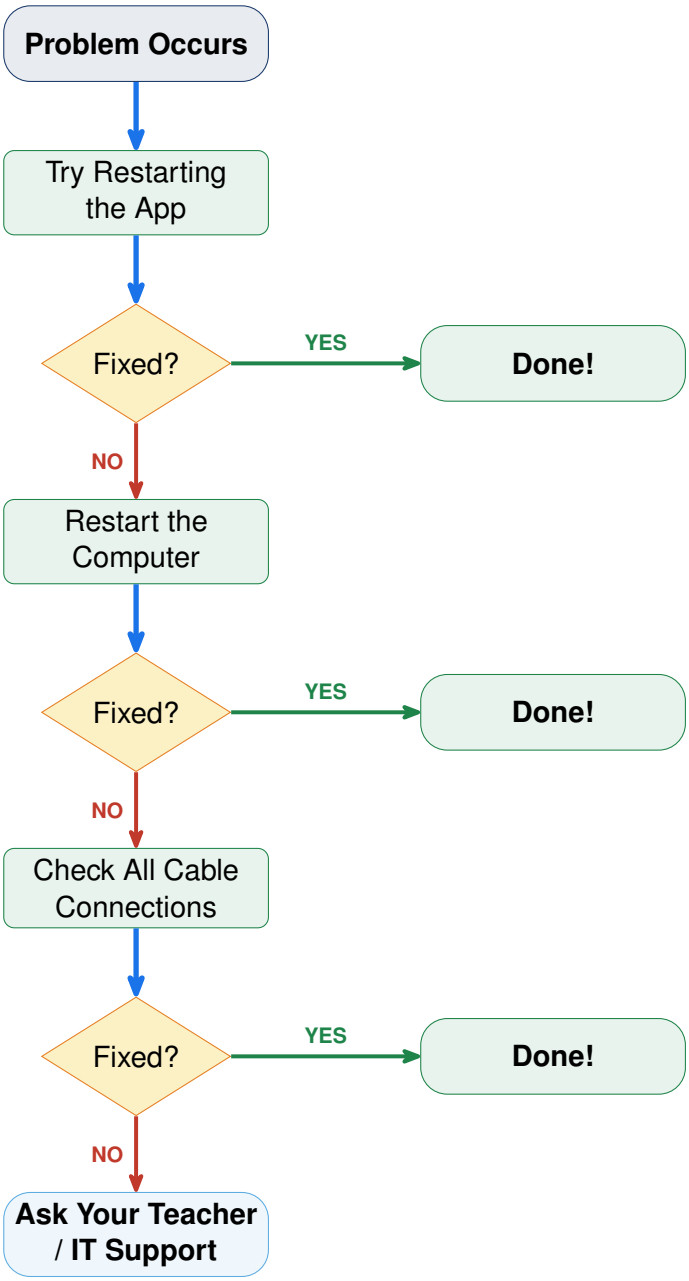


Figure 16.1: The troubleshooting flowchart — follow each step before escalating

16.2 Common Problems and Fixes

Table 16.1: Common computer problems and quick fixes

Problem	Possible Cause	Quick Fix
Screen is black	Monitor off or cable loose	Check power & cables
No sound	Volume muted or low	Click speaker icon, unmute
App frozen	Program crashed	Ctrl + Alt + Del → Task Manager
No internet	Wi-Fi disconnected	Reconnect to Wi-Fi
Slow computer	Too many apps open	Close unused programs

If your screen is completely frozen and nothing responds, hold the Power button for 5–10 seconds to force restart. But only do this as a **last resort** — it can cause data loss!

Keyboard Shortcuts

Ctrl + Alt + Del	Open security options / Task Manager
Ctrl + Shift + Esc	Open Task Manager directly
Alt + F4	Close the current window/program

Did You Know?

The Ctrl + Alt + Del shortcut was invented by **David Bradley**, an IBM engineer, in 1981. He originally designed it as a quick way to restart during development. He once joked: "I may have invented it, but Bill Gates made it famous!"

Review Questions

Q1. What is the first thing you should try when an app stops responding?

Q2. How do you open Task Manager?

Q3. Name three common computer problems and their fixes.

Q4. When should you ask your teacher for help instead of trying to fix it yourself?

Q5. What does Ctrl + Alt + Del do?

Challenge Task

Open **Task Manager** (`Ctrl`+`Shift`+`Esc`). Look at the **Processes** tab. Which program is using the most memory? Which is using the most CPU? Write down your findings!

17 Part II — Hands-On Practicals

Time to put your Windows skills to the test! These practicals cover everything from navigating the desktop to managing files like a pro.

Practical 1: Exploring the Windows Desktop 10 min

You Will Need: Computer 10 min Observation sheet

Objective: Identify and interact with all desktop elements.

Steps:

1. Identify the Start button, Taskbar, System Tray, and desktop icons.
2. Click the Start button and explore the menu.
3. Right-click on the desktop and choose “Personalise.”
4. Find the current time and date in the System Tray.
5. Write down the names of all pinned apps on the Taskbar.

Expected Outcome: Students identify all desktop elements correctly.

Practical 2: Changing Desktop Wallpaper 10 min

You Will Need: Computer 10 min

Objective: Personalise the desktop appearance.

Steps:

1. Right-click on an empty area of the desktop.
2. Select “Personalise” from the context menu.
3. Click “Background” and choose a new wallpaper from the list.
4. Try selecting “Solid colour” and pick your favourite colour.
5. Change it back to the original wallpaper when finished.

Expected Outcome: Students successfully change and restore the wallpaper.

Practical 3: File and Folder Operations 15 min

You Will Need: Computer 15 min File Explorer

Objective: Create, rename, copy, move, and delete files and folders.

Steps:

1. Open **File Explorer** (`Windows`+`E`).
2. Navigate to the **Documents** folder.

3. Create a new folder: Right-click → New → Folder. Name it “Practice.”
4. Inside “Practice,” create two more folders: “Text” and “Images.”
5. Right-click inside “Text” → New → Text Document. Name it “Hello.txt.”
6. Right-click “Hello.txt” → Copy. Navigate to “Images” → Paste.
7. Rename the copy to “HelloCopy.txt” (select it, press `F2`).
8. Delete “HelloCopy.txt” (press `Delete`).
9. Check the Recycle Bin — is the file there?

Expected Outcome: Students complete all file operations successfully.

Practical 4: Using Notepad 10 min

You Will Need: Computer 10 min Keyboard

Objective: Create, edit, and save a text file using Notepad.

Steps:

1. Open Notepad (Start → search “Notepad”).
2. Type: “My name is [your name]. I am learning about Windows.”
3. Press `Enter` and type today’s date.
4. Save the file: `Ctrl`+`S` → Save in Documents as “MyNote.txt.”

5. Close Notepad. Reopen “MyNote.txt” to verify your text is saved.

Expected Outcome: Students save and retrieve a text file.

Practical 5: Drawing in Paint 15 min

You Will Need: Computer 15 min Paint

Objective: Use Paint’s drawing tools to create a simple picture.

Steps:

1. Open MS Paint (Start → search “Paint”).
2. Use the **Rectangle** and **Triangle** shapes to draw a house.
3. Use the **Circle** tool to draw a sun.
4. Add colour using the **Fill** tool (paint bucket).
5. Use the **Text** tool to write your name at the bottom.
6. Save as “MyDrawing.png” in Documents (`Ctrl`+`S`).

Expected Outcome: A colourful drawing saved as a PNG file.

Practical 6: Using the Calculator 10 min

You Will Need: Computer 10 min Calculator

Objective: Perform calculations using the Windows Calculator.

Steps:

1. Open Calculator (Start → search "Calculator").
2. Calculate: 456 × 78. Write down the answer.
3. Calculate: 9999 ÷ 3. Write down the answer.
4. Switch to **Scientific** mode (click the menu icon).
5. Calculate 5^3 (type 5, then use the x^y button, type 3, press =).
6. Switch to **Programmer** mode and explore binary numbers!

Expected Outcome: Students use Standard and Scientific calculator modes.

Practical 7: Checking System Information 10 min

You Will Need: Computer 10 min Worksheet

Objective: Find and record basic system information.

Steps:

1. Press `Windows`+`I` to open Settings.
2. Go to **System** → **About**.
3. Write down: Computer name, Processor, RAM amount, Windows version.
4. Open **This PC** and check total storage on the C: drive.
5. Record the total and free storage space.

Expected Outcome: Students record their computer's specifications.

Practical 8: Taking Screenshots 10 min

You Will Need: Computer 10 min Snipping Tool

Objective: Capture and save screenshots using different methods.

Steps:

1. Press `PrtSc` to capture the full screen. Open Paint → `Ctrl`+`V` to paste.
2. Save as "FullScreen.png."
3. Press `Windows`+`Shift`+`S` to open Snipping Tool / Snip & Sketch.
4. Select a rectangular area of the screen.
5. The snip is copied to clipbord — paste it into Paint and save as "MySnip.png."

Expected Outcome: Students capture and save two types of screenshots.

Scan for extra Part II resources and activities!

PART III

Computer Basics and Shortcuts

Go deeper into hardware, software, and master the keyboard shortcuts that make you lightning fast.

CPU RAM Storage Shortcuts Mouse Skills

Part III

Computer Basics and Shortcuts

18 Advanced Computer Concepts

Learning Objectives

- Explore how computers are used across industries
- Understand the Input–Process–Output cycle in depth
- Appreciate the impact of computers on modern society
- Recognise emerging technologies and their possibilities

You already know the basics — now let's crank it up a notch! In this chapter, we explore how computers are transforming *every* part of the real world.

18.1 Computers in the Real World

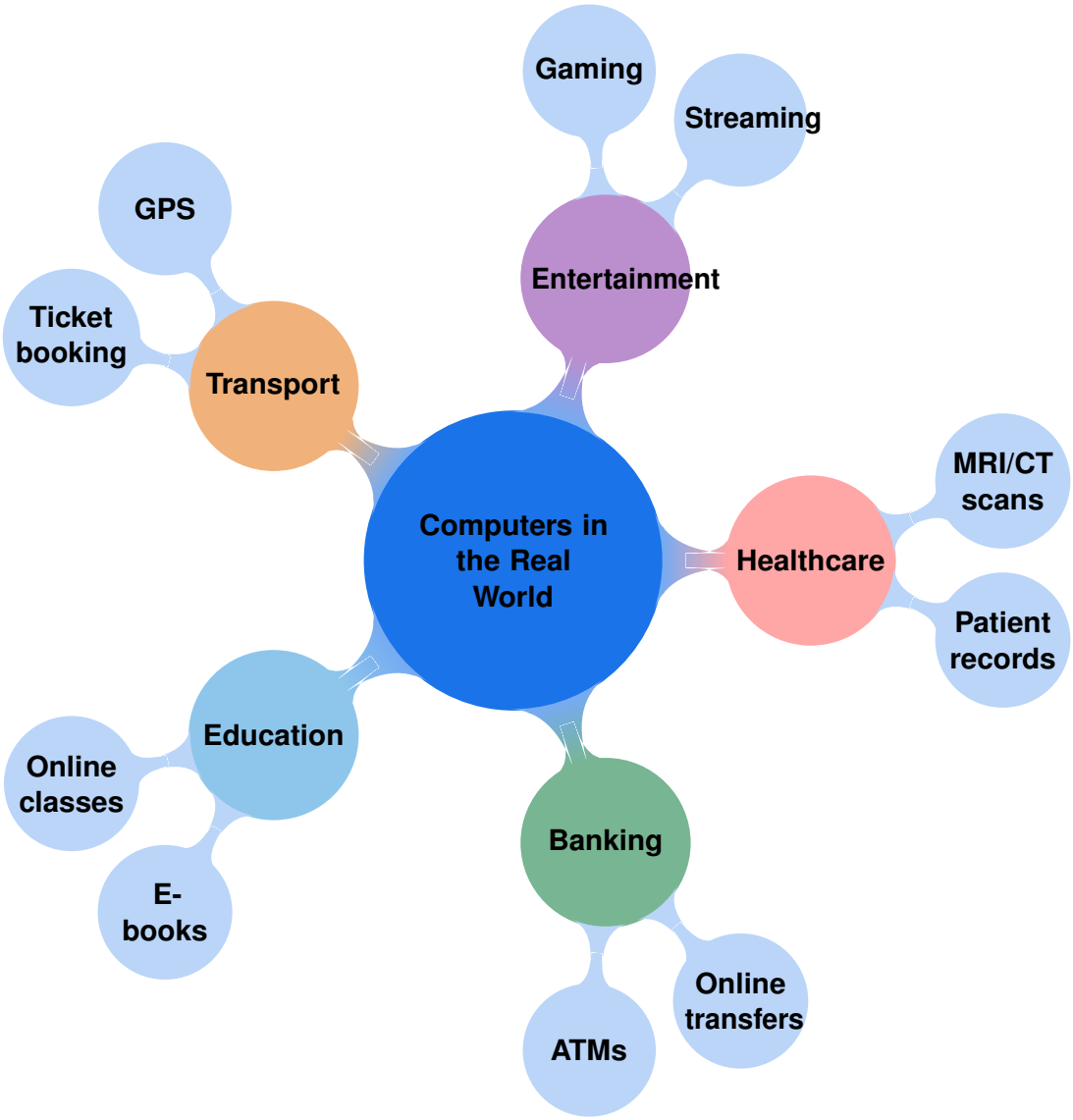


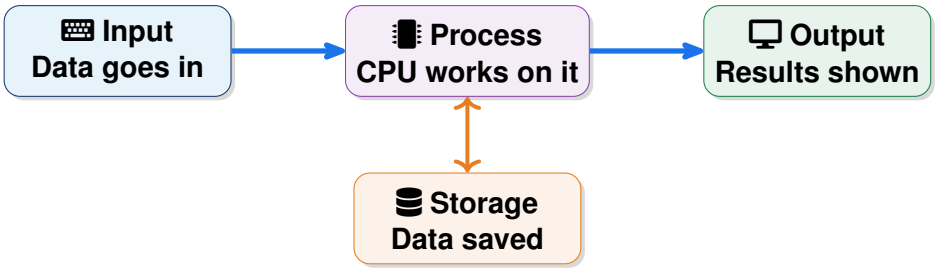
Figure 18.1: Computers are used in virtually every industry

Table 18.1: Computer applications across different domains

Domain	Application	Software/Device
Education	Online classes, digital textbooks	Zoom, Google Classroom
Banking	ATM transactions, online banking	Banking apps, UPI
Healthcare	Patient records, diagnostic imaging	Hospital management systems
Entertainment	Video streaming, gaming	Netflix, Steam
Transport	GPS navigation, ticket booking	Google Maps, IRCTC

18.2 The IPO Cycle Revisited

Every computer task follows the **Input–Process–Output** (IPO) cycle:



Everything you do on a computer follows the IPO cycle. Typing a letter? Input = keystrokes, Process = converting to text, Output = letters on screen, Storage = saving the document!

Real World Connection

When you search on Google: **Input** = your search query, **Process** = Google's servers scanning billions of pages, **Output** = the search results you see. This entire process takes less than one second!

Did You Know?

There are approximately **5.5 billion** internet users worldwide — that's about 68% of the entire human population! And the number is growing every day.

Review Questions

Q1. Name five industries where computers are widely used.

Q2. What is the IPO cycle? Give an example.

Q3. How are computers used in healthcare?

Q4. What role do computers play in education today?

Q5. Name two emerging technologies that use computers.

Challenge Task

Choose any **one industry** (e.g., agriculture, space, sports). Research how computers are used in that field and write a 10-sentence paragraph explaining what you found. Include at least one specific software or device name.

19 Hardware in Detail

Learning Objectives

- Understand the internal components of the CPU
- Explain the difference between RAM and ROM
- Describe how the ALU and Control Unit work together
- Compare different types of memory and storage

Time to look *inside* the computer! We're going to open up the CPU and see what's really going on in there. Don't worry — no screwdrivers needed!

19.1 Inside the CPU

The CPU has two main parts that work together:

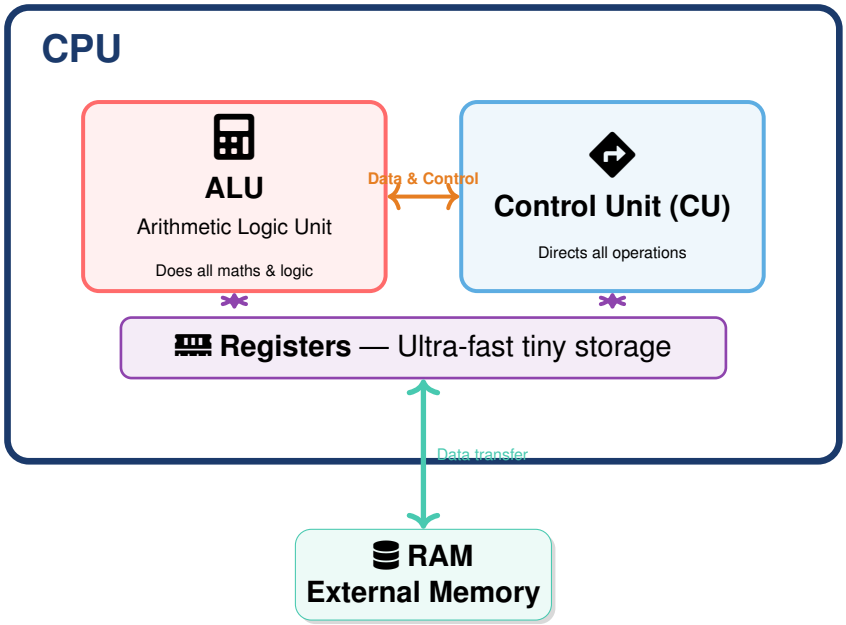


Figure 19.1: Inside the CPU: ALU, Control Unit, and Registers

19.2 RAM vs ROM

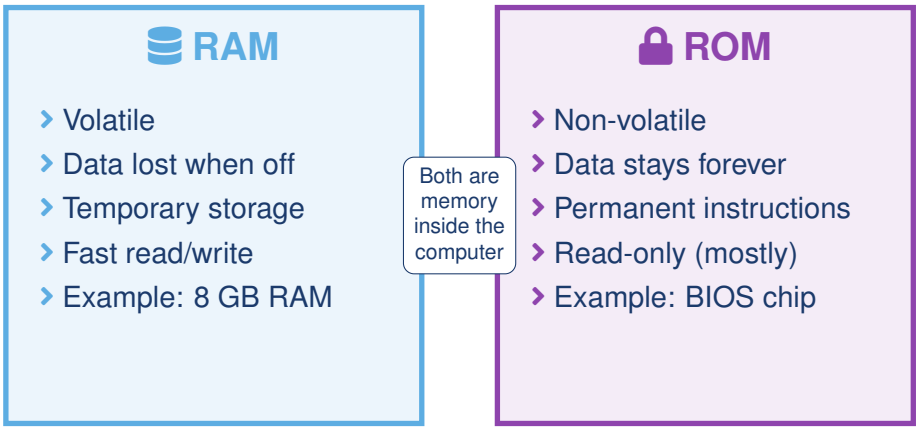


Figure 19.2: RAM vs ROM — two types of memory with very different jobs

RAM vs ROM Quick Summary

	RAM	ROM
Full name	Random Access Memory	Read-Only Memory
Volatile?	Yes (loses data)	No (keeps data)
Speed	Very fast	Slower than RAM
Used for	Running programs	Boot instructions

Here's an easy way to remember: **RAM** is like your desk — it holds what you're currently working on. **ROM** is like a recipe book — the information is permanent and doesn't change.

Did You Know?

The first computers had only a few **kilobytes** of RAM. Today, even budget smartphones have 4–6 GB of RAM — that's over a million times more! Modern gaming PCs can have 64 GB or even 128 GB of RAM.

Review Questions

Q1. What are the two main parts inside the CPU?

Q2. What does ALU stand for and what does it do?

Q3. What is the difference between RAM and ROM?

Q4. What are registers in a CPU?

Q5. Why is RAM called "volatile" memory?

Challenge Task

Check your school computer's RAM: press `Windows` + `I` → System → About. How much RAM does it have? Research online: how much RAM does a typical smartphone have? Compare the two and write three sentences about your findings.

20 Software in Detail

Learning Objectives

- Classify software into system and application categories
- Understand the role of device drivers
- Explore different types of application software
- Learn about open-source vs proprietary software

We covered software basics earlier — now it's time to go deeper! There are many more types of software than you might think.

20.1 System Software Revisited

System software includes more than just the operating system:

Types of System Software

1. **Operating System** — Manages hardware and provides user interface (Windows, macOS)
2. **Device Drivers** — Small programs that help the OS communicate with specific hardware (printer driver, graphics driver)
3. **Utility Software** — Tools for maintenance (antivirus, disk cleanup, file compression)
4. **Firmware** — Permanent software built into hardware (BIOS, router firmware)

20.2 Application Software Categories

Table 20.1: Types of application software

Category	Purpose	Example
Word Processor	Create and edit documents	MS Word, Google Docs
Spreadsheet	Organise data and calculations	MS Excel, Google Sheets
Presentation	Create slideshows	MS PowerPoint
Web Browser	Access websites	Chrome, Edge, Firefox
Media Player	Play audio and video	VLC, Windows Media Player
Image Editor	Edit photos and images	Paint, Photoshop, GIMP

20.3 Open Source vs Proprietary

- **Open Source** — Free to use, modify, and share. The code is public. Examples: Linux, Firefox, GIMP, LibreOffice.
- **Proprietary** — Owned by a company. You must pay (or get a licence) to use it. Code is secret. Examples: Windows, MS Office, Photoshop.

Did you know that **Android** is based on the open-source **Linux** kernel? That means the core of your smartphone's OS was created by volunteers around the world, for free!

Did You Know?

LibreOffice is a free, open-source alternative to Microsoft Office. It includes a word processor, spreadsheet, and presentation tool — and it's used by millions of people, governments, and schools worldwide who don't want to pay for MS Office!

Review Questions

Q1. What are device drivers and why are they needed?

Q2. Name three types of system software besides the OS.

Q3. What is the difference between open-source and proprietary software?

Q4. Give two examples each of open-source and proprietary software.

Q5. What category of application software would you use to edit photos?

Challenge Task

Research **three free, open-source** alternatives to paid software. For example, find a free alternative to Photoshop, MS Word, and Windows. Make a comparison table with columns: Paid Software, Free Alternative, Key Difference.

21 Operating Systems in Depth

Learning Objectives

- Compare different operating systems (Windows, macOS, Linux, Android, iOS)
- Understand multitasking and how the OS manages multiple programs
- Learn about user accounts and permissions
- Explore the concept of a graphical user interface (GUI)

You already know *what* an OS does. Now let's understand *how* it does it — and explore the world beyond Windows!

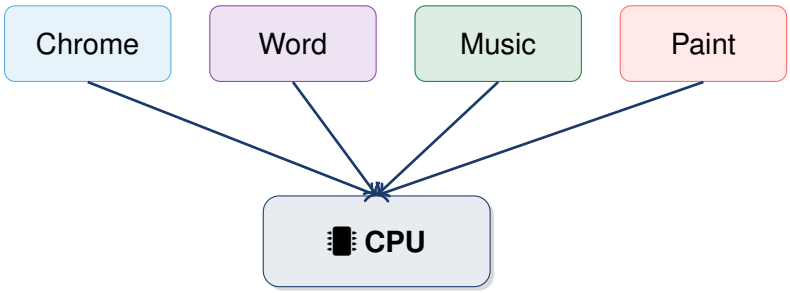
21.1 Comparing Operating Systems

Table 21.1: Popular operating systems compared

OS	Made By	Used On	Cost
Windows	Microsoft	Desktops, laptops	Paid
macOS	Apple	MacBooks, iMacs	Free (with Mac)
Linux	Community	Servers, desktops	Free
Android	Google	Smartphones, tablets	Free
iOS	Apple	iPhones, iPads	Free (with device)
ChromeOS	Google	Chromebooks	Free (with device)

21.2 Multitasking

Multitasking means running multiple programs at the same time. The OS manages this by rapidly switching between programs — so fast that it feels like everything is running simultaneously.



CPU switches between all programs incredibly fast

21.3 GUI — Graphical User Interface

A GUI (Graphical User Interface) lets you interact with the computer using visual elements — icons, buttons, windows, and menus — instead of typing text commands. All modern operating systems use a GUI.

Before GUIs existed, people used a **CLI** (Command Line Interface) — a black screen where you typed commands. It looked like something from a hacker movie! Some IT professionals still prefer CLI because it's faster for certain tasks.

Did You Know?

The first GUI was developed at **Xerox PARC** in 1973 — yes, the same Xerox that makes photocopiers! Steve Jobs visited their lab, was inspired, and created the **Apple Lisa** and later the **Macintosh** (1984), which brought GUIs to regular people.

Review Questions

Q1. Name four popular operating systems and who makes each one.

Q2. What is multitasking? How does the CPU handle it?

Q3. What does GUI stand for and why is it important?

Q4. Which operating system is free and open-source?

Q5. What is the difference between GUI and CLI?

Challenge Task

If you have access to a tablet or smartphone, compare its interface to a Windows desktop. List **five differences** and **five similarities** between the two operating systems' interfaces.

22 Advanced File Management

Learning Objectives

- Understand file extensions and what they mean
- Use advanced search to find files quickly
- Manage file properties and attributes
- Compress (zip) and extract files

You learned the basics of files in Part II. Now let's become file management **experts** — you'll learn tricks that even many adults don't know!

22.1 File Extensions

Every file has an extension — the part after the dot in the filename. It tells the computer what type of file it is and which program should open it.

Table 22.1: Common file extensions and their meanings

Extension	Type	Description	Opens With
.txt	Text	Plain text file	Notepad
.docx	Document	Word document	MS Word
.xlsx	Spreadsheet	Excel workbook	MS Excel
.pptx	Presentation	PowerPoint slides	MS PowerPoint
.pdf	Document	Portable document	PDF reader
.jpg / .png	Image	Photo or picture	Photos / Paint
.mp3	Audio	Music file	Media Player
.mp4	Video	Video file	Media Player
.zip	Archive	Compressed file(s)	File Explorer

22.2 Searching for Files

Lost a file? Use Windows Search:

1. Press Windows + S or click the search bar.

2. Type part of the file name.
3. Filter by type: Documents, Photos, Videos, etc.
4. Double-click the result to open it.

22.3 Compressing (Zipping) Files

Compressing a file makes it smaller, which is great for sharing via email or USB.

How to Zip a File

1. Right-click the file or folder you want to compress.
2. Select "Compress to ZIP file" (or "Send to → Compressed folder").
3. A new `.zip` file appears in the same location.
4. To **extract**: right-click the zip file → "Extract All."

Changing a file's extension (e.g., renaming `.docx` to `.txt`) doesn't convert the file — it just confuses the computer about how to open it! Use "Save As" to properly convert file types.

Did You Know?

The ZIP file format was invented by **Phil Katz** in 1989. The name "zip" was chosen to imply speed — the idea was that zipping files together would be as fast as a zipper closing on a jacket!

Review Questions

Q1. What is a file extension, and why is it important?

Q2. Name the file extensions for: a Word document, an image, a video.

Q3. How do you search for a file on Windows?

Q4. What does "compressing" a file do?

Q5. Why should you NOT manually change a file's extension?

Challenge Task

Create a folder called "ZipTest" with three files inside (any types). Right-click the folder and compress it into a ZIP file. Check the file sizes: how much smaller is

the ZIP compared to the original folder? Express it as a percentage!

23 Keyboard and Mouse Skills

Learning Objectives

- Identify and name key groups on a keyboard
- Use the mouse confidently for all basic actions
- Understand ergonomic typing posture
- Practise touch-typing techniques

The keyboard and mouse are your main tools for communicating with the computer. Mastering them is like learning to ride a bicycle — once you've got it, you've got it for life!

23.1 The Keyboard Layout

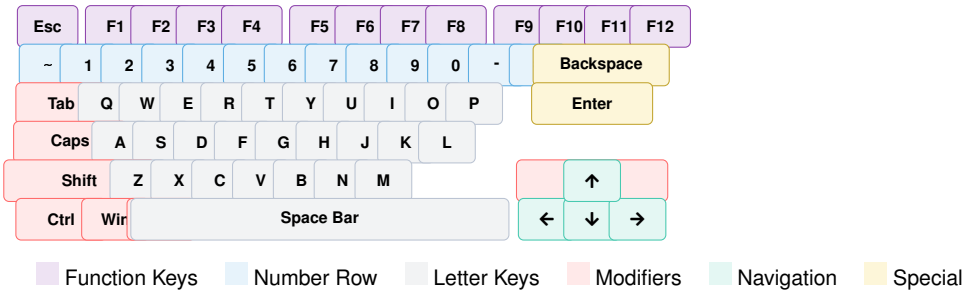


Figure 23.1: A colour-coded keyboard map showing key groups

23.2 Mouse Anatomy and Actions

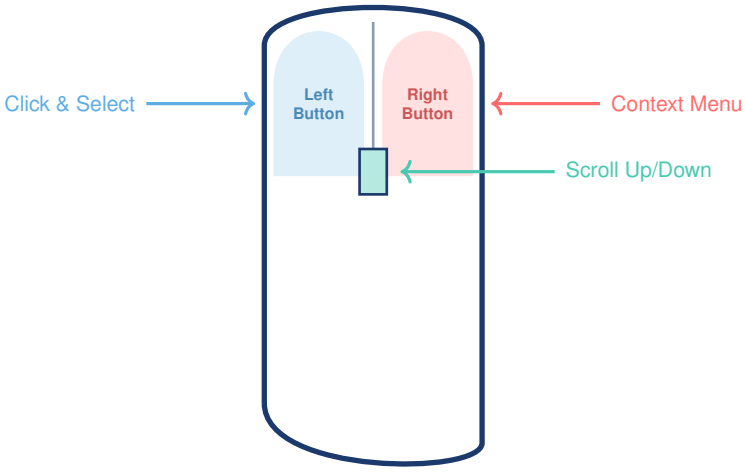


Figure 23.2: Mouse anatomy — learn what each part does

Table 23.1: Mouse actions and what they do

Action	How To Do It	What Happens
Single Click	Press left button once	Selects item
Double Click	Press left button twice quickly	Opens item
Right Click	Press right button once	Opens context menu
Click & Drag	Hold left, move mouse	Moves or selects area
Scroll	Roll scroll wheel	Page moves up or down

BYTE

The first computer mouse was invented in 1964 by **Douglas Engelbart**. It was made of wood and had only one button! It was called a "mouse" because the cable coming out of the back looked like a tail.

Did You Know?

Modern optical mice use a tiny camera that takes up to **18,000 pictures per second** of the surface below to detect movement. That's faster than any video camera you've ever seen!

Review Questions

Q1. Name and describe five key groups on a keyboard.

Q2. What is the difference between single-click and double-click?

Q3. What key do you press to type capital letters temporarily?

Q4. Name the three main parts of a mouse.

Q5. What does the scroll wheel do?

Challenge Task

Time yourself typing a paragraph of 50 words using proper finger placement (don't look at the keyboard!). Record how long it took. Practise for a week and time yourself again. How much did you improve?

24 Keyboard Shortcuts

Learning Objectives

- Memorise the most important keyboard shortcuts
- Use shortcuts to work faster than using a mouse
- Apply shortcuts in Word, Excel, and general Windows tasks
- Understand why shortcuts improve productivity

Keyboard shortcuts are like **superpowers** for your fingers. Once you learn them, you'll wonder how you ever survived without them!

24.1 Why Use Shortcuts?

Using the mouse for everything is slow. Keyboard shortcuts let you do the same things *much* faster — sometimes 10 times faster!

24.2 Essential Shortcuts Cheat Sheet

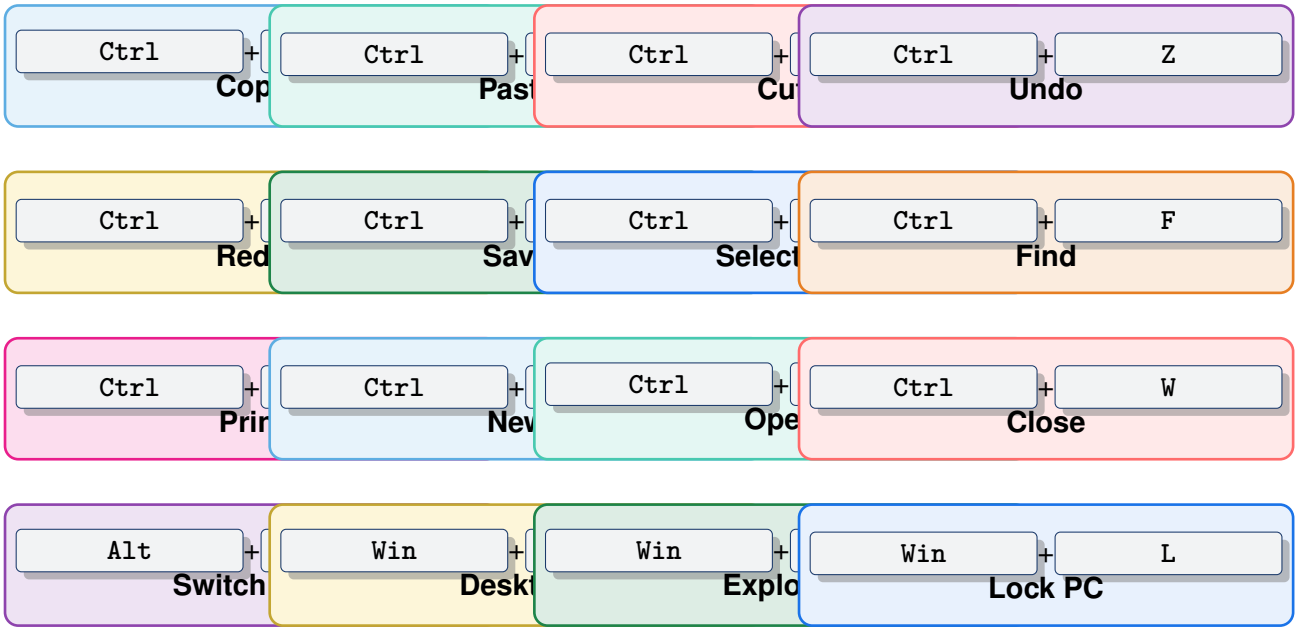


Figure 24.1: The 16 most important keyboard shortcuts — learn them all!

24.3 Shortcuts Save Time

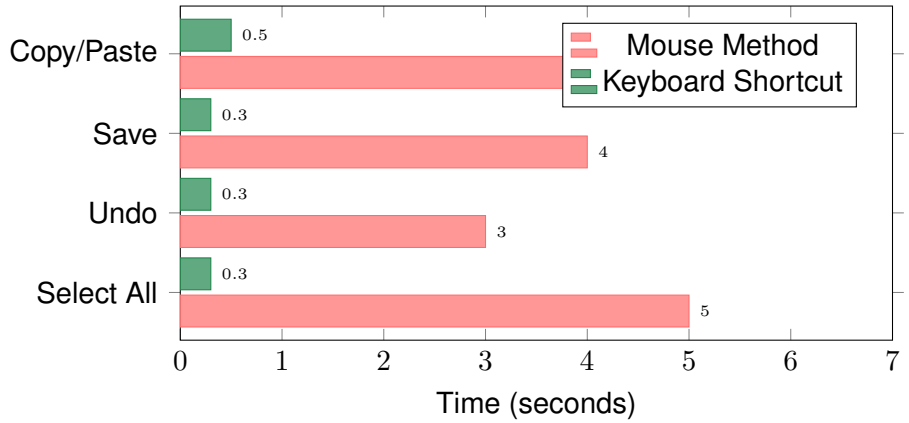


Figure 24.2: Shortcuts are up to 10x faster than using the mouse!

Professional typists use shortcuts *hundreds* of times per day. Over a year, that saves an estimated **8 full working days** of time! That's more than a week of extra holiday, just from pressing `Ctrl`+`S` instead of clicking menus.

Keyboard Shortcuts

Pro tip: The most important shortcut to build as a habit is `Ctrl`+`S` (Save). Press it every few minutes while working — it only takes a fraction of a second and could save you from losing hours of work if the power goes out!

Try It Yourself!

Open Notepad and type a sentence. Now practise these shortcuts in order:

1. `Ctrl`+`A` (select all text)
2. `Ctrl`+`C` (copy it)
3. Press `Enter` twice, then `Ctrl`+`V` (paste it)
4. `Ctrl`+`Z` (undo the paste)
5. `Ctrl`+`S` (save the file)

You just used 5 shortcuts in under 10 seconds!

Did You Know?

The `Ctrl`+`C`, `Ctrl`+`X`, `Ctrl`+`V` shortcuts were popularised by Apple's original Macintosh in 1984 (using the Command key instead of Ctrl). They chose C for Copy, X for Cut (like scissors cutting), and V for Paste (because V looks like an

insertion mark in proofreading).

Review Questions

Q1. What are keyboard shortcuts, and why should you use them?

Q2. Write the shortcuts for: Copy, Paste, Cut, Undo, Save.

Q3. What does `Alt`+`Tab` do?

Q4. What does `Win`+`E` open?

Q5. How much time can shortcuts save compared to using the mouse?

Challenge Task

Create a "Shortcuts Study Card": on one side, write the shortcut (e.g., `Ctrl`+`C`). On the other side, write what it does ("Copy"). Make at least 12 cards and quiz a partner!

25 Part III — Hands-On Practicals

These practicals will sharpen your hardware knowledge, file skills, and make you a keyboard shortcut ninja! Let's go!

Practical 1: Identifying Internal Hardware 10 min

You Will Need: Computer or diagram 10 min Worksheet

Objective: Identify CPU, RAM, and storage components.

Steps:

1. Look at the diagram your teacher provides showing the inside of a computer.
2. Label the CPU, RAM sticks, hard drive/SSD, motherboard, and power supply.
3. For each component, write one sentence about what it does.
4. Compare RAM and ROM — fill in a comparison table on your worksheet.
5. Open Settings → About and check your computer's CPU and RAM specs.

Expected Outcome: Students correctly identify and describe 5 internal components.

Practical 2: Exploring File Extensions 10 min

You Will Need: Computer 10 min File Explorer

Objective: Identify file types by their extensions.

Steps:

1. Open File Explorer and navigate to Documents.
2. In the View menu, enable “File name extensions” so you can see them.
3. Find at least 5 different file types and record their extensions.
4. For each file, note the extension and what program opens it.
5. Create a new text file and try changing its extension to .docx — what happens?

Expected Outcome: Students identify 5+ file extensions and their associated programs.

Practical 3: Zipping and Unzipping Files 10 min

You Will Need: Computer 10 min Files to compress

Objective: Compress files into a ZIP archive and extract them.

Steps:

1. Create a folder called “ZipPractice” in Documents.

2. Inside it, create 3 text files with different names and some content.
3. Right-click the folder → "Compress to ZIP file."
4. Note the size of the original folder and the ZIP file. Is the ZIP smaller?
5. Right-click the ZIP file → "Extract All" to a new location.
6. Verify all 3 files are intact in the extracted folder.

Expected Outcome: Students compress and extract files successfully.

Practical 4: Keyboard Typing Practice 15 min

You Will Need: Computer 15 min Keyboard

Objective: Practise typing with correct finger placement.

Steps:

1. Open Notepad.
2. Place your fingers on the **home row**: left hand on A-S-D-F, right on J-K-L-;
3. Type the following without looking at the keyboard: "the quick brown fox jumps over the lazy dog"
4. Now type it again, timing yourself. Record your time.
5. Type it three more times, trying to get faster each time.

6. Write down your fastest time.

Expected Outcome: Students improve their typing speed over 5 attempts.

Practical 5: Keyboard Shortcuts Practice 15 min

You Will Need: Computer 15 min Keyboard

Objective: Use 10 different keyboard shortcuts in a real task.

Steps:

1. Open **Notepad** (`Win` → type "Notepad" → `Enter`).
2. Type three sentences about your favourite hobby.
3. Use `Ctrl`+`A` to select all text.
4. Use `Ctrl`+`C` to copy, then `Ctrl`+`V` to paste it below.
5. Use `Ctrl`+`Z` to undo the paste.
6. Use `Ctrl`+`S` to save as "ShortcutsPractice.txt."
7. Use `Ctrl`+`F` to find the word "favourite."
8. Use `Alt`+`Tab` to switch to File Explorer.
9. Use `Win`+`D` to show the desktop.
10. Use `Win`+`E` to open File Explorer again.

Expected Outcome: Students use 10 shortcuts confidently.

Practical 6: Mouse Skills Challenge 10 min

You Will Need: Computer 10 min Mouse

Objective: Master all mouse actions in a timed challenge.

Steps:

1. **Single click:** Select 5 different desktop icons (one at a time).
2. **Double-click:** Open Recycle Bin, then close it.
3. **Right-click:** Right-click the desktop, explore the menu, press Esc.
4. **Drag and drop:** Rearrange 3 desktop icons to new positions.
5. **Scroll:** Open a long document or web page and scroll to the bottom.
6. Time yourself doing all 5 tasks. Record your total time.

Expected Outcome: Students complete all mouse actions within 3 minutes.

Scan for extra Part III resources and activities!

PART IV

Microsoft Word

Master the art of creating professional documents — from simple notes to beautifully formatted reports.

Documents Formatting Tables Images Printing

Part IV

Microsoft Word

26 Introduction to MS Word

Learning Objectives

- Open MS Word and identify parts of the Word window
- Create, save, and open a document
- Understand the Ribbon, Quick Access Toolbar, and Status Bar
- Use basic text entry and editing features

MS Word is the world's most popular word processor. Whether it's a school essay, a letter, or a poster — Word can handle it all. Let's get started!

26.1 The MS Word Window

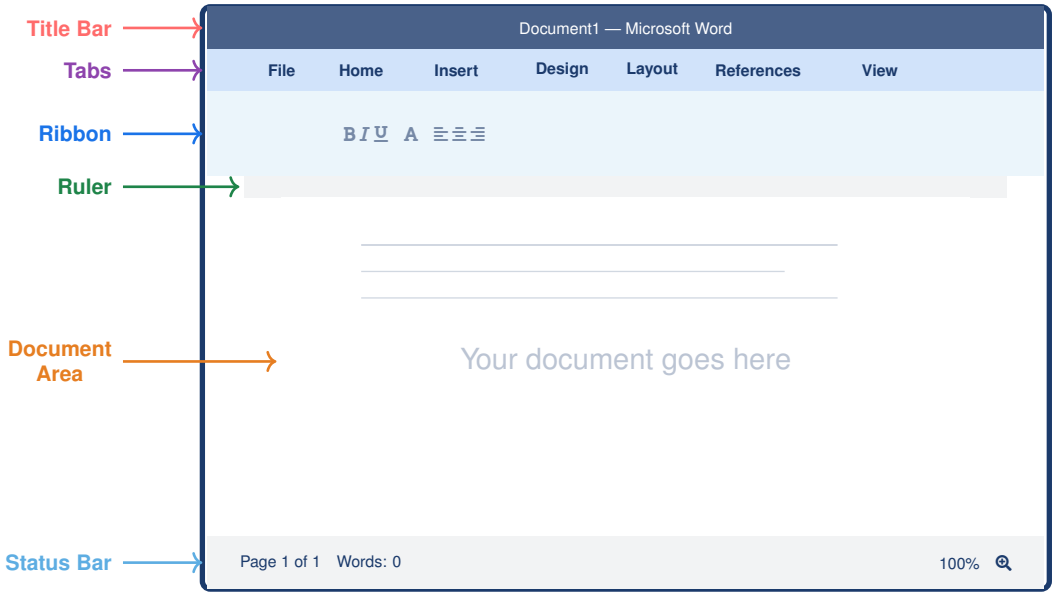


Figure 26.1: The MS Word window with all parts labelled

26.2 Creating and Saving Documents

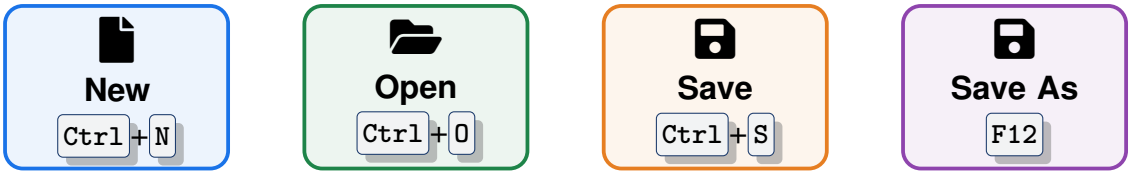


Figure 26.2: Four essential file operations in MS Word

Note

Save (`Ctrl`+`S`) saves changes to the current file. **Save As** (`F12`) lets you save a copy with a different name or in a different location. Use Save As when you want to keep the original AND create a modified version.

Get into the habit of pressing `Ctrl`+`S` every few minutes! There's nothing worse than losing 30 minutes of work because the power went out or the computer froze.

Did You Know?

Microsoft Word was first released in 1983 as "Multi-Tool Word" for the Xerox ALTO. Its original price was **$375** (about $1,100 in today's money). Today, you can use Word for free online at office.com!

Review Questions

Q1. Name six parts of the MS Word window.

Q2. What is the difference between Save and Save As?

Q3. What keyboard shortcut opens a new document?

Q4. Where is the Ribbon located and what does it contain?

Q5. How do you open an existing Word document?

Challenge Task

Open MS Word and create a document about yourself. Include your name, school, class, hobbies, and favourite subject. Save it as "AboutMe.docx" in your Documents folder. Then use **Save As** to save another copy called "AboutMe_Backup.docx."

27 Editing Text in MS Word

Learning Objectives

- Select, copy, cut, paste, and move text efficiently
- Use Find and Replace to edit large documents
- Undo and redo actions
- Use spell check and autocorrect features

BYTE

Editing is where the real magic happens. You rarely get text perfect on the first try — and that's fine! Word's editing tools make fixing things super easy.

27.1 Selecting Text

Selection Techniques

To select...	Do this...
A word	Double-click the word
A line	Click in the left margin next to the line
A sentence	Hold `Ctrl` and click anywhere in the sentence
A paragraph	Triple-click anywhere in the paragraph
All text	Press `Ctrl`+`A`
Custom range	Click at start, hold `Shift`, click at end

27.2 Copy, Cut, and Paste

Keyboard Shortcuts

`Ctrl`+`C`	Copy	Copies selected text to clipboard
`Ctrl`+`X`	Cut	Removes text and copies to clipboard
`Ctrl`+`V`	Paste	Inserts text from clipboard
`Ctrl`+`Z`	Undo	Reverses the last action
`Ctrl`+`Y`	Redo	Re-applies an undone action

27.3 Find and Replace

Need to change "colour" to "color" throughout a 20-page document? Don't do it manually!

Using Find and Replace

1. Press `Ctrl`+`H` to open Find and Replace.
2. In "Find what," type the word you want to find.
3. In "Replace with," type the new word.
4. Click "Replace All" to change every occurrence at once.
5. A message tells you how many replacements were made.

27.4 Spell Check

Word automatically underlines spelling errors with a red wavy line and grammar errors with a blue wavy line. Right-click the underlined word to see suggestions.

Word's spell checker knows over **100,000 words** in English! But it can't catch everything — if you type "their" when you mean "there," it won't notice because both are real words. Always proofread your work!

Did You Know?

The clipboard (where copied text goes) was named after a real **clipboard** — the board with a clip that office workers used to hold papers. The computer version was invented at Xerox PARC in the 1970s.

Review Questions

Q1. How do you select an entire paragraph in Word?

Q2. What is the difference between Cut and Copy?

Q3. What shortcut opens Find and Replace?

Q4. What does a red wavy underline mean in Word?

Q5. How do you undo the last action?

Challenge Task

Type a paragraph of at least 6 sentences with **intentional spelling mistakes**. Use Word's spell checker to find and fix all of them. Then use Find and Replace to change one word throughout the paragraph. Count how many replacements were made.

28 Formatting in MS Word

Learning Objectives

- Apply font formatting: bold, italic, underline, font size, font colour
- Change text alignment: left, centre, right, justify
- Adjust line spacing and paragraph spacing
- Use bullet points and numbered lists

Formatting is what turns a boring block of text into a beautiful, readable document. Think of it like decorating a room — same space, totally different vibe!

28.1 Font Formatting

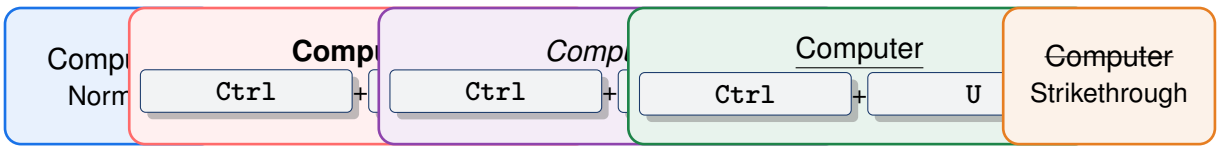


Figure 28.1: Different font formatting styles applied to the same word

Keyboard Shortcuts

Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + Shift + >	Increase font size
Ctrl + Shift + <	Decrease font size

28.2 Text Alignment

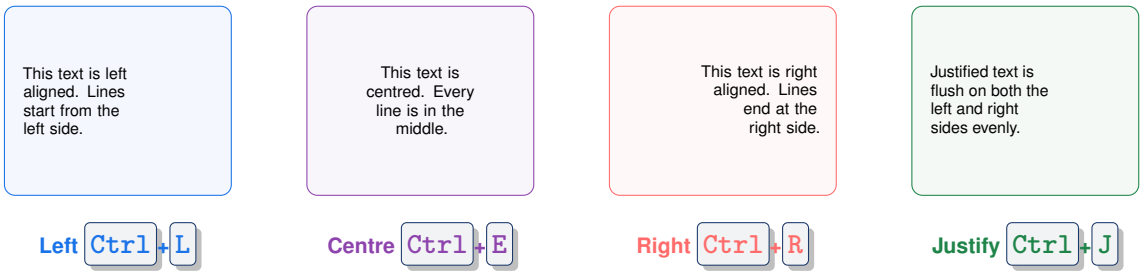


Figure 28.2: Four text alignment options in MS Word

28.3 Line Spacing

Table 28.1: Line spacing options and when to use them

Spacing	Value	When To Use
Single	1.0	Compact notes, short documents
1.5 Lines	1.5	Most school assignments (recommended)
Double	2.0	Essays the teacher will annotate

28.4 Lists

Word supports two types of lists:

- **Bulleted lists** — for items with no specific order (like this one!)
- **Numbered lists** — for steps or items in a specific order

To create a list: Home tab → click the Bullets ☰ or Numbering ⁱ☰ button.

The word "font" comes from the French word "fonte," meaning "something that has been melted." This is because early printing used melted metal to create letter shapes. Now we just click a dropdown menu — much less dangerous!

Try It Yourself!

Open a new Word document and type a title "My Favourite Things." Make the title **bold**, *size 18*, and blue. Below it, create a bulleted list of 5 things you love. Then change the alignment of the title to **centre**. Finally, set the line spacing to 1.5.

Did You Know?

There are over **200,000 fonts** available in the world! Windows comes with about 200 pre-installed. The most commonly used fonts for school work are **Calibri**, **Arial**, and **Times New Roman**.

Review Questions

Q1. Name four types of font formatting you can apply in Word.

Q2. What are the four text alignment options and their shortcuts?

Q3. What line spacing is recommended for school assignments?

Q4. How do you create a bulleted list in Word?

Q5. What is the shortcut for bold?

Challenge Task

Create a **one-page formatted document** about your school. It must include: a centred bold title (size 20), a subtitle in italic, two paragraphs with justified alignment, a bulleted list of school subjects, and at least 3 different font colours. Save it as “MySchool.docx.”

29 Inserting Elements

Learning Objectives

- Insert and format tables in a document
- Add pictures, shapes, and WordArt
- Insert headers, footers, and page numbers
- Use the Insert tab effectively

Plain text is fine, but adding tables, images, and shapes makes your document look **professional**. Let's learn how to spice things up!

29.1 Inserting Tables

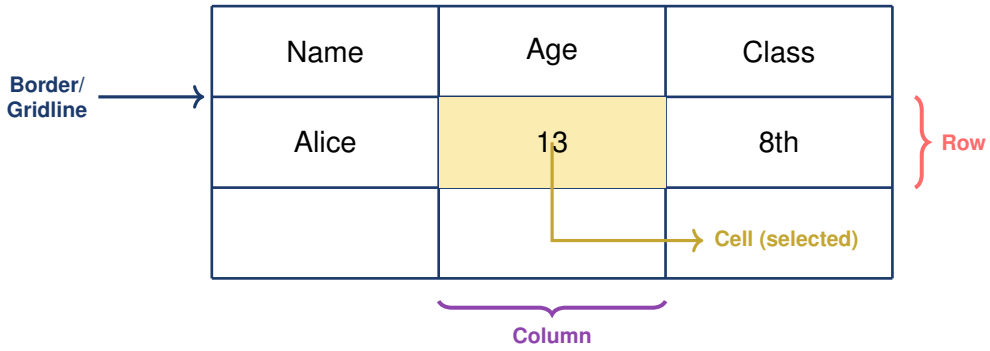


Figure 29.1: Anatomy of a table: rows, columns, cells, and borders

To insert a table: **Insert** tab → **Table** → drag to select the number of rows and columns.

29.2 Inserting Pictures and Shapes

Inserting a Picture

1. Click where you want the picture to appear.
2. Go to **Insert** tab → **Pictures**.

3. Choose “This Device” to insert from your computer, or “Online Pictures” for web images.
4. Select the image and click **Insert**.
5. Resize by dragging the corner handles. Use the **Layout Options** button to choose how text wraps around the image.

Figure 29.2: The Insert tab groups — your gateway to adding elements

29.3 Headers, Footers, and Page Numbers

Headers and Footers

- A **Header** appears at the **top** of every page (e.g., document title, your name)
- A **Footer** appears at the **bottom** of every page (e.g., page number, date)
- To add them: **Insert** tab → **Header** or **Footer** or **Page Number**

BYTE

You can insert **WordArt** to create fancy decorated text for titles and headings. Go to Insert → WordArt and choose a style. It’s like having a graphic designer inside your computer!

Did You Know?

The WordArt feature was introduced in **Word 97** and became an instant hit. Students in the late ’90s used it on literally everything — to the point where teachers had to *ban* excessive WordArt usage in assignments!

Review Questions

Q1. How do you insert a table in MS Word?

Q2. What are the parts of a table called?

Q3. How do you insert a picture from your computer?

Q4. What is the difference between a header and a footer?

Q5. How do you add page numbers to a document?

Challenge Task

Create a document with: (a) A WordArt title, (b) A 3-column, 5-row table of your classmates' names, favourite colours, and hobbies, (c) An inserted picture, (d) Page numbers in the footer. Save as "FormattedDoc.docx."

30 Page Layout and Printing

Learning Objectives

- Change page orientation between portrait and landscape
- Set page margins and paper size
- Preview and print a document
- Add page borders and watermarks

Before you print, you need to make sure your page looks right on paper. Layout settings control how your document will appear when printed.

30.1 Portrait vs Landscape

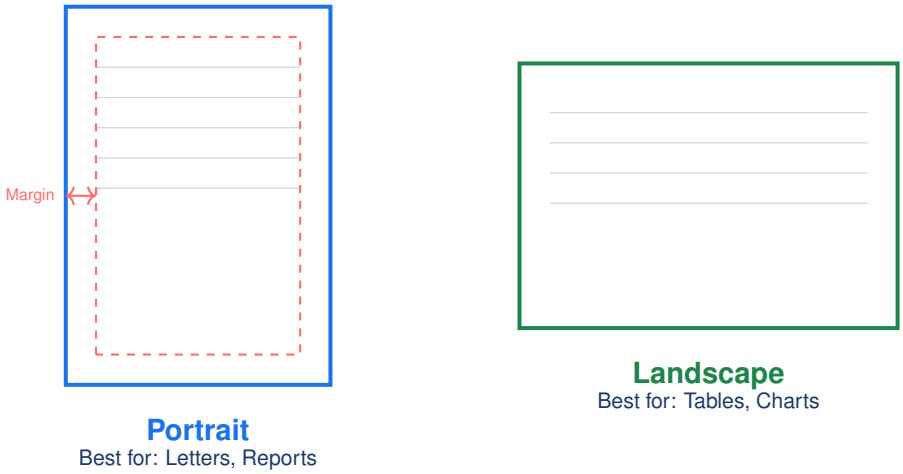


Figure 30.1: Portrait (tall) vs Landscape (wide) orientation

To change orientation: **Layout** tab → **Orientation** → choose Portrait or Landscape.

30.2 Margins

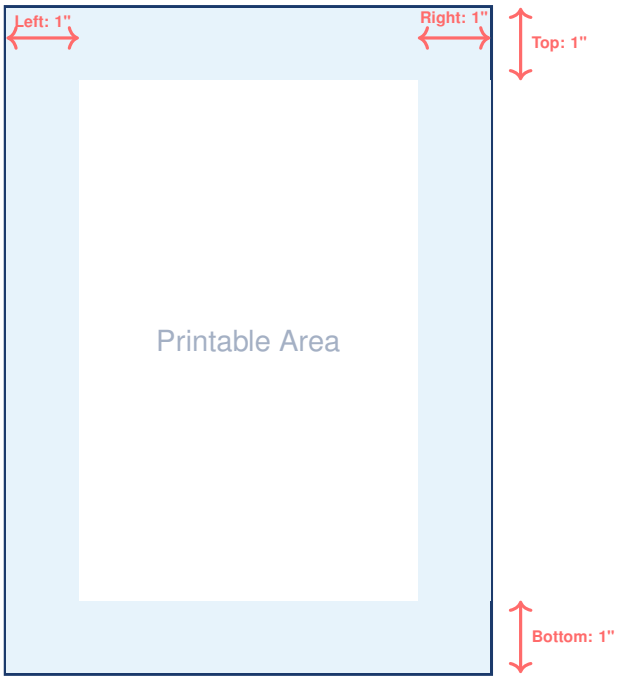


Figure 30.2: Page margins define the space between text and the edge of the paper

30.3 Printing a Document

How to Print

1. Press `Ctrl`+`P` or go to File → Print.
2. The **Print Preview** shows how your document will look on paper.
3. Choose the number of copies.
4. Select your printer from the list.
5. Click **Print**.

Note

Always check the **Print Preview** before printing! This prevents wasting paper on a document that doesn't look right. Think green — save trees! 🌲

Before printing, make sure a printer is connected and turned on. If you see "Printer Offline," check the cable, Wi-Fi connection, or ask your teacher for help.

Did You Know?

The average office worker uses about **10,000 sheets** of paper per year. That's equivalent to one whole tree! Using Print Preview and printing only what you need helps save the environment.

Review Questions

Q1. What is the difference between Portrait and Landscape?

Q2. How do you change page margins in Word?

Q3. What keyboard shortcut opens the Print dialog?

Q4. Why should you always use Print Preview before printing?

Q5. What are margins, and why are they important?

Challenge Task

Create a document with: Portrait orientation, "Moderate" margins, a page border (Design tab → Page Borders), and your name in the header. Open Print Preview and take a screenshot of how it looks. Save as "LayoutPractice.docx."

31 Part IV — Hands-On Practicals

BYTE

Time to become a Word wizard! These practicals take you from basic typing all the way to creating beautifully formatted documents.

Practical 1: Creating Your First Document 10 min

You Will Need: Computer 10 min MS Word

Objective: Open Word, type content, and save a document.

Steps:

1. Open MS Word. Click “Blank document.”
2. Type a 5-sentence paragraph about your favourite animal.
3. Save the document as “MyFirstDoc.docx” (`Ctrl`+`S`).
4. Close Word. Reopen Word and open your saved file (`Ctrl`+`O`).
5. Verify your text is still there.

Expected Outcome: Students create and save their first Word document.

Practical 2: Text Selection and Editing 10 min

You Will Need: Computer 10 min MS Word

Objective: Select, copy, cut, paste, and rearrange text.

Steps:

1. Open "MyFirstDoc.docx" from Practical 1.
2. Double-click a word to select it. Press `Ctrl`+`B` to bold it.
3. Select the entire first sentence (`Ctrl`+click). Copy it (`Ctrl`+`C`).
4. Place cursor at the end. Paste (`Ctrl`+`V`).
5. Use `Ctrl`+`Z` to undo. Use `Ctrl`+`H` to find and replace one word.

Expected Outcome: Students edit text using shortcuts confidently.

Practical 3: Paragraph Formatting 15 min

You Will Need: Computer 15 min MS Word

Objective: Apply alignment, spacing, and indentation.

Steps:

1. Create a new document with a title and 3 paragraphs.
2. Centre-align the title (`Ctrl`+`E`).

3. Justify the paragraphs (`Ctrl` + `J`).
4. Change line spacing to 1.5 (Home → Line Spacing).
5. Add first-line indent to each paragraph (Layout → Indent).
6. Save as "FormattedParagraphs.docx."

Expected Outcome: A professionally formatted 3-paragraph document.

Practical 4: Text Formatting Showcase 15 min

You Will Need: Computer 15 min MS Word

Objective: Apply font formatting to create a visually appealing document.

Steps:

1. Type a title: "My Formatting Showcase." Make it **bold**, size 22, colour blue.
2. Type a subtitle in *italic*, size 14, colour grey.
3. Type a paragraph and apply: different font (Arial), size 12, justified.
4. Highlight one important phrase using the highlight tool (yellow).
5. Create a bulleted list of 4 items and a numbered list of 4 steps.
6. Save as "FormatShowcase.docx."

Expected Outcome: A document demonstrating 5+ formatting techniques.

Practical 5: Working with Tables 15 min

You Will Need: Computer 15 min MS Word

Objective: Insert, format, and populate a table.

Steps:

1. Create a new document. Type a title: "Class Timetable."
2. Insert a 6-column, 7-row table (Insert → Table).
3. Fill in: Column headers = Day, Period 1–5. Rows = Monday–Friday + subjects.
4. Bold the header row and shade it blue (Table Design tab).
5. Apply alternating row colours (Table Design → Banded Rows).
6. Save as "Timetable.docx."

Expected Outcome: A colour-formatted class timetable.

Practical 6: Inserting Pictures and Shapes 15 min

You Will Need: Computer 15 min MS Word

Objective: Add visual elements to a document.

Steps:

1. Create a new document. Add a WordArt title at the top.
2. Insert a picture from your computer (Insert → Pictures).
3. Resize the picture and set text wrapping to "Square."
4. Insert a shape (Insert → Shapes) and add text inside it.
5. Insert a table below the picture with 3 columns and 4 rows.
6. Save as "VisualDoc.docx."

Expected Outcome: A document with WordArt, picture, shape, and table.

Practical 7: Headers, Footers, and Page Numbers 10 min

You Will Need: Computer 10 min MS Word

Objective: Add headers, footers, and automatic page numbers.

Steps:

1. Open any document you created in previous practicals.
2. Insert a **header**: Insert → Header → type your name.
3. Insert a **footer**: Insert → Footer → type "Confidential."
4. Insert page numbers: Insert → Page Number → Bottom of Page.
5. Double-click outside the header/footer area to return to the document.
6. Save the file.

Expected Outcome: Document with header, footer, and page numbers on every page.

Practical 8: Complete Document Project 20 min

You Will Need: Computer 20 min MS Word

Objective: Create a multi-page, fully formatted document.

Steps:

1. Create a 2-page document about “My Favourite Festival.”
2. Page 1: WordArt title, introductory paragraph (justified, 1.5 spacing), inserted picture.
3. Page 2: Table with 3 columns (Activity, Time, Description), numbered list of traditions.
4. Add header with your name, footer with page numbers.
5. Apply a consistent colour theme throughout.
6. Save as “FestivalDoc.docx” and check Print Preview (`Ctrl`+`P`).

Expected Outcome: A professional 2-page document using all skills learned.

Scan for extra Part IV resources and activities!

PART V

Microsoft Excel

Unlock the power of spreadsheets — organise data, crunch numbers, and create stunning charts.

Spreadsheets Cells Formulas Charts Sorting

Part V

Microsoft Excel

32 Introduction to MS Excel

Learning Objectives

- Open MS Excel and identify parts of the Excel window
- Understand cells, rows, columns, and cell addresses
- Enter data into cells (text, numbers, dates)
- Navigate a worksheet using keyboard and mouse

Excel is like a super-powered calculator combined with a smart notebook. It can do maths, organise lists, create charts, and much more. Let's explore!

32.1 The Excel Window

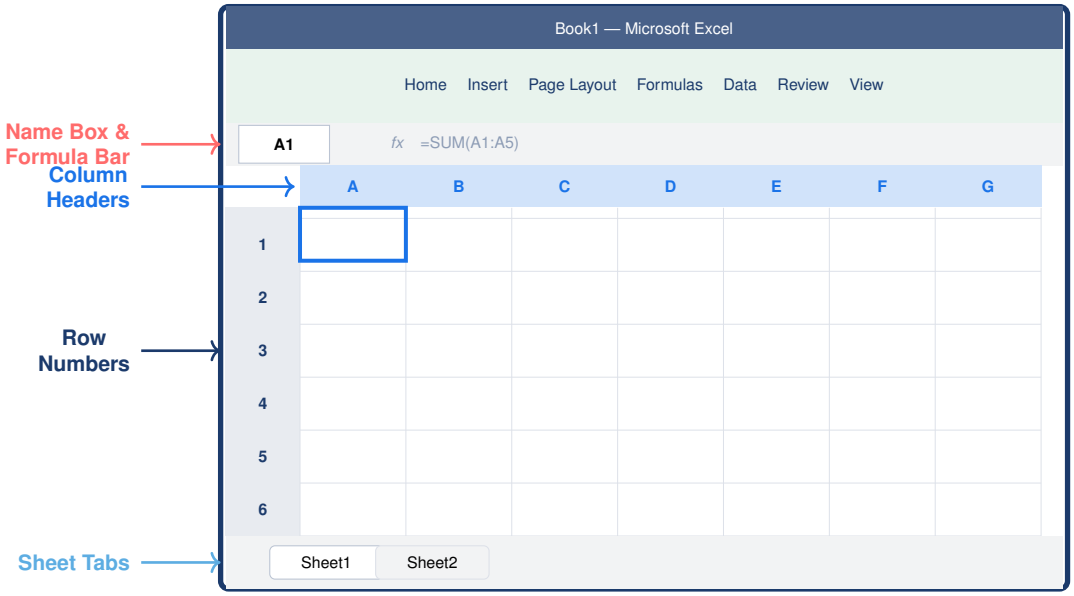


Figure 32.1: The MS Excel window with all key elements labelled

32.2 Understanding Cell Addresses

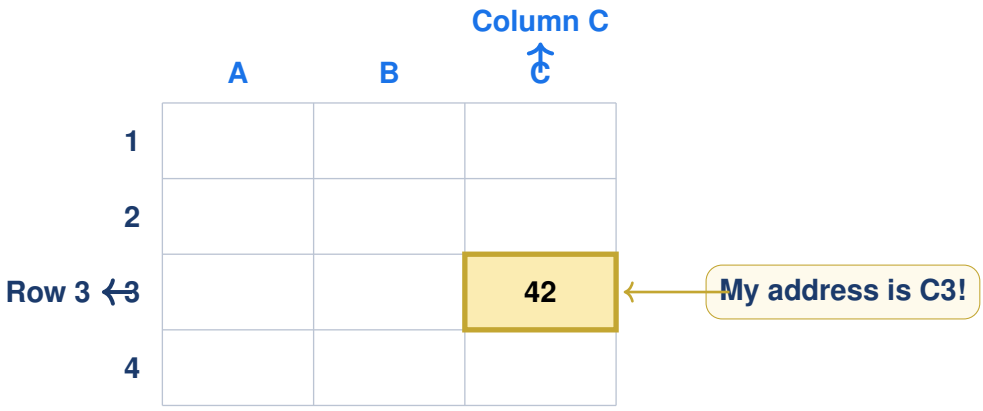


Figure 32.2: Cell C3 — column letter + row number = cell address

Cell Address

Every cell in Excel has a unique address made of its **column letter** and **row number**. For example, the cell in column C, row 3 has the address **C3**. This is how Excel knows which cell you're talking about in formulas.

An Excel worksheet has **16,384 columns** (A to XFD) and **1,048,576 rows**. That's over **17 billion cells** in a single sheet! You'd need to live several lifetimes to fill them all.

Did You Know?

The first spreadsheet program was called **VisiCalc**, released in 1979 for the Apple II. It was so popular that people bought Apple II computers *just* to run VisiCalc. It's considered one of the most important software programs ever created!

Review Questions

Q1. Name five parts of the Excel window.

Q2. What is a cell address? Give three examples.

Q3. How many columns and rows does an Excel worksheet have?

Q4. Where is the Formula Bar, and what does it show?

Q5. What are sheet tabs, and what do they do?

Challenge Task

Open Excel and type data for 5 classmates: Name (column A), Age (column B), Favourite Subject (column C). Bold the header row. Adjust column widths so all text is visible. Save as “ClassData.xlsx.”

33 Working with Worksheets

Learning Objectives

- Add, rename, delete, and rearrange worksheets
- Enter different types of data (text, numbers, dates)
- Adjust column widths and row heights
- Use AutoFill to quickly enter data series

BYTE

A workbook is like a notebook with multiple pages. Each page is a **worksheet** (or "sheet"). You can have as many sheets as you need!

33.1 Managing Worksheets

Worksheet Operations

Action	How To Do It
Add a new sheet	Click the **+** button next to sheet tabs
Rename a sheet	Double-click the sheet tab and type a new name
Delete a sheet	Right-click the tab → Delete
Move a sheet	Click and drag the tab to a new position
Copy a sheet	Right-click → Move or Copy → check "Create a copy"

33.2 Entering Data

Excel cells can hold three types of data:

- **Text** (labels) — aligned to the left by default (e.g., "Name", "Total")
- **Numbers** — aligned to the right by default (e.g., 42, 3.14, -5)
- **Dates** — displayed in date format (e.g., 13/04/2026)

33.3 AutoFill

AutoFill is one of Excel's most powerful features. It automatically fills cells with a pattern you start.

Using AutoFill

1. Type **1** in cell A1 and **2** in cell A2.
2. Select both cells (A1:A2).
3. Hover over the small square at the bottom-right corner (the **fill handle**).
4. Click and drag down to A10.
5. Excel automatically fills: 3, 4, 5, 6, 7, 8, 9, 10!

AutoFill also works with: days of the week, months, dates, and custom patterns.

AutoFill is smart enough to continue patterns like "Mon, Tue, Wed..." or "Jan, Feb, Mar..." It even works with "Quarter 1, Quarter 2, Quarter 3..." — try it!

Did You Know?

Excel's grid has been the same size since Excel 2007: 16,384 columns and 1,048,576 rows. Before that (Excel 2003), it only had 256 columns and 65,536 rows. The upgrade was a big deal for scientists and data analysts!

Review Questions

Q1. How do you rename a worksheet?

Q2. What three types of data can you enter in a cell?

Q3. What is AutoFill and how do you use it?

Q4. How do you add a new worksheet to a workbook?

Q5. How can you adjust the width of a column?

Challenge Task

Create a workbook with 3 sheets: "Names", "Marks", and "Summary." In "Names," use AutoFill to create a numbered list (1–20) in column A. In column B, add 20 names. Rename all three sheet tabs with meaningful names.

34 Formatting in Excel

Learning Objectives

- Format cells: font, colour, borders, alignment
- Apply number formats (currency, percentage, date)
- Use cell styles and conditional formatting
- Merge cells for headers

A well-formatted spreadsheet is SO much easier to read than a plain one. Let's make your data look professional!

34.1 Cell Formatting

You can format cells just like text in Word — change font, size, colour, bold, italic, and alignment. Select cells first, then use the Home tab.

Keyboard Shortcuts

Ctrl + B	Bold selected cells
Ctrl + 1	Open Format Cells dialog
Ctrl + Shift + $	Apply Currency format
Ctrl + Shift + %	Apply Percentage format

34.2 Number Formats

Table 34.1: Common number formats in Excel

Format	Example	Use For
General	1234.5	Default, no specific format
Number	1,234.50	Numbers with decimal places
Currency	$1,234.50	Money and prices
Percentage	85%	Ratios and proportions
Date	13/04/2026	Calendar dates
Text	001234	Text that looks like numbers (PINs, codes)

34.3 Merging Cells

Merge & Centre combines multiple cells into one large cell. It's perfect for creating headers that span multiple columns.

How to merge: Select the cells → Home tab → **Merge & Centre** button.

Warning!

When you merge cells, only the data in the **top-left cell** is kept. All other data in the selected range is **deleted**. Always merge *before* typing your header, or make sure only one cell has data.

Excel can display numbers in many ways. The number 0.85 can be shown as "0.85," "85%," or even "85/100" (fraction format). It's the same number — just different outfits!

Did You Know?

The most popular font in Excel is **Calibri** (11pt) — the same default font used in Word. Before 2007, the default was **Arial** (10pt). The change happened because Calibri is specifically designed for on-screen reading.

Review Questions

Q1. How do you apply bold formatting to a cell in Excel?

Q2. Name four number formats available in Excel.

Q3. What does Merge & Centre do?

Q4. What shortcut opens the Format Cells dialog?

Q5. Why should you be careful when merging cells that contain data?

Challenge Task

Create a “Monthly Budget” spreadsheet with columns: Item, Cost, Category. Enter at least 8 items. Format the Cost column as **Currency**. Bold and colour the header row. Add borders to all cells. Merge cells A1:C1 for a title.

35 Formulas and Functions

Learning Objectives

- Write basic formulas using +, -, *, /
- Use functions: SUM, AVERAGE, MIN, MAX, COUNT
- Understand cell references in formulas
- Use the fill handle to copy formulas

This is where Excel truly shines! Formulas and functions turn your spreadsheet from a simple table into a powerful calculator. Let's unlock this superpower!

35.1 Writing Formulas

Every formula in Excel starts with an **equals sign** (=).

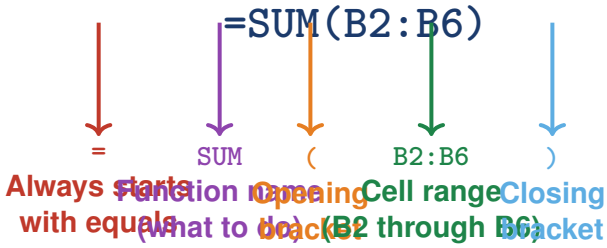


Figure 35.1: Anatomy of a formula — every part explained

35.2 Essential Functions

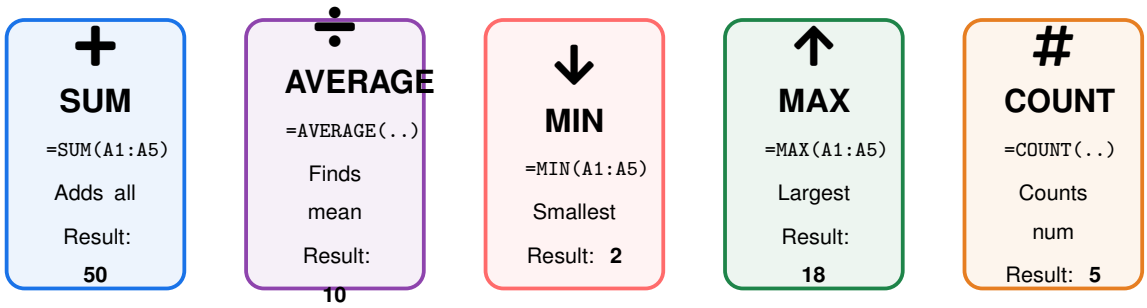


Figure 35.2: Five essential Excel functions you must know

Table 35.1: Common Excel functions reference

Function	What It Does	Example	Result
SUM	Adds all values in a range	`=SUM(A1:A5)`	50
AVERAGE	Finds the mean (average)	`=AVERAGE(A1:A5)`	10
MIN	Finds the smallest value	`=MIN(A1:A5)`	2
MAX	Finds the largest value	`=MAX(A1:A5)`	18
COUNT	Counts cells with numbers	`=COUNT(A1:A5)`	5

35.3 The Fill Handle

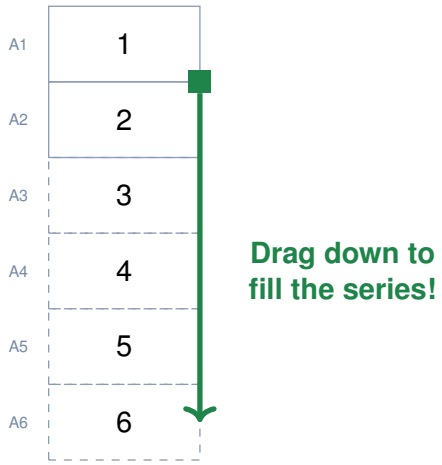


Figure 35.3: The fill handle copies formulas and extends data patterns

Remember: formulas always start with =. If you type `SUM(A1:A5)` without the equals sign, Excel treats it as text, not a formula! You'll see the text "SUM(A1:A5)" instead of the result.

Excel can handle formulas with up to **8,192 characters** and **64 levels of nested functions**. Some finance professionals write formulas so complex they look like alien code!

Did You Know?

The most commonly used function in Excel is **SUM**, followed by AVERAGE and VLOOKUP. Together, these three functions are used in over **90%** of all Excel spreadsheets worldwide!

Review Questions

Q1. What symbol must every formula start with?

Q2. Write the formula to add cells B1 through B10.

Q3. What does the AVERAGE function do?

Q4. What is the difference between MIN and MAX?

Q5. What is the fill handle and how do you use it?

Challenge Task

Create a "Test Scores" spreadsheet with 5 students and 3 subjects. In column E, use `=SUM` to calculate each student's total. In row 8, use `=AVERAGE` for each subject's average. In row 9, use `=MAX` for the highest score. In row 10, use `=MIN` for the lowest.

36 Charts and Data Management

Learning Objectives

- Create bar charts, line charts, and pie charts from data
- Sort data in ascending and descending order
- Use basic filters to find specific data
- Choose the right chart type for your data

Charts turn boring numbers into beautiful visuals. A single chart can tell a story that would take paragraphs of text. Let's learn to create stunning charts!

36.1 Types of Charts

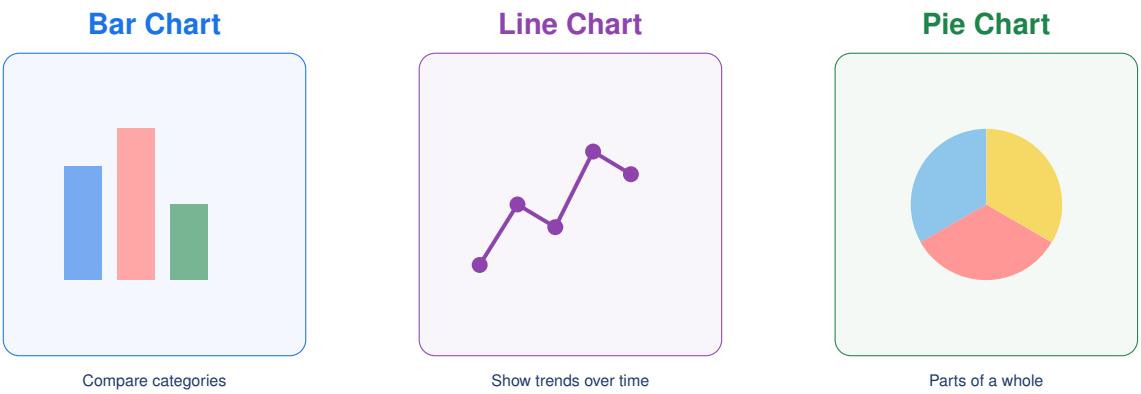


Figure 36.1: Three chart types and when to use each one

Creating a Chart

1. Select the data you want to chart (including headers).
2. Go to **Insert** tab → **Charts** group.
3. Choose your chart type (Bar, Line, or Pie).
4. Click the chart to customise: add a title, labels, and colours.

5 Resize and move the chart as needed.

36.2 Sorting Data

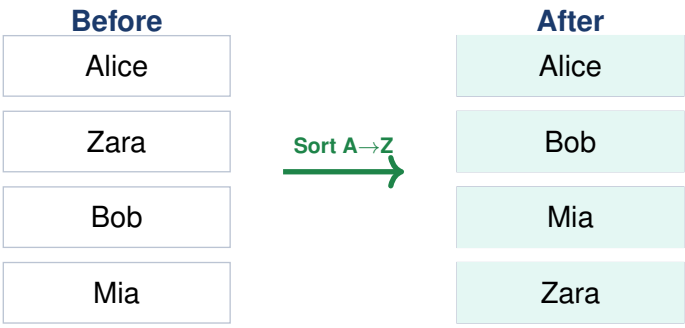


Figure 36.2: Sorting rearranges data in alphabetical or numerical order

To sort: Select your data → **Data** tab → click **Sort A to Z** or **Sort Z to A**.

BYTE

Excel can create over **20 different chart types**, including radar charts, waterfall charts, treemaps, and even 3D surface plots! For school, you'll mostly use bar, line, and pie charts.

Did You Know?

The first pie chart was created in 1801 by **William Playfair**, a Scottish engineer. He used it to show the proportions of the Turkish Empire's territory in Asia, Europe, and Africa. Pie charts are over 200 years old!

Review Questions

Q1. Name three types of charts you can create in Excel.

Q2. When would you use a pie chart instead of a bar chart?

Q3. What are the steps to create a chart from your data?

Q4. How do you sort data alphabetically?

Q5. What is a filter and how does it help?

Challenge Task

Enter marks for 5 students in 4 subjects. Create a **bar chart** comparing all students' total marks. Then create a **pie chart** showing how one student's marks are distributed across subjects. Add chart titles and labels.

37 Part V — Hands-On Practicals

BYTE

Spreadsheet time! These practicals will take you from entering your first number to creating charts that would impress your maths teacher.

Practical 1: Exploring the Excel Interface 10 min

You Will Need: Computer 10 min MS Excel

Objective: Identify all parts of the Excel window.

Steps:

1. Open Excel and create a new blank workbook.
2. Click on cell A1. Notice the cell address in the Name Box.
3. Click on cell D5. What does the Name Box show now?
4. Look at the Formula Bar — what does it display?
5. Find the Sheet Tabs at the bottom. How many sheets are there?

Expected Outcome: Students identify Name Box, Formula Bar, and Sheet Tabs.

Practical 2: Entering and Formatting Data 15 min

You Will Need: Computer 15 min MS Excel

Objective: Enter data and apply basic formatting.

Steps:

1. In A1, type "Student Name." In B1, type "Maths." In C1, type "Science."
2. Enter 5 students' names and marks in the rows below.
3. Bold the header row. Change the header background to blue with white text.
4. Apply borders to all data cells (Home → Borders).
5. Adjust column widths by double-clicking the column border.
6. Save as "StudentMarks.xlsx."

Expected Outcome: A formatted table with 5 students' marks.

Practical 3: Using AutoFill 10 min

You Will Need: Computer 10 min MS Excel

Objective: Use AutoFill to create data series.

Steps:

1. In A1 type "Monday." Use AutoFill to fill A1:A7 with days of the week.

2. In B1 type “January.” Use AutoFill to fill B1:B12 with months.
3. In C1 type 2, in C2 type 4. Select both and AutoFill to C10 (even numbers).
4. In D1 type “Quarter 1.” AutoFill down to D4.
5. Observe how Excel recognises each pattern!

Expected Outcome: Students fill 4 different series using AutoFill.

Practical 4: Writing Basic Formulas 15 min

You Will Need: Computer 15 min MS Excel

Objective: Create formulas using +, -, *, /.

Steps:

1. Open “StudentMarks.xlsx” from Practical 2.
2. In D1 type “Total.” In D2 type `=B2+C2` and press Enter.
3. Use the fill handle to copy the formula down for all 5 students.
4. In E1 type “Average.” In E2 type `=(B2+C2)/2` and fill down.
5. Click on D2 and check the formula in the Formula Bar.
6. Save the file.

Expected Outcome: Students calculate totals and averages using formulas.

Practical 5: Using Functions (SUM, AVERAGE, MAX, MIN) 15 min

You Will Need: Computer 15 min MS Excel

Objective: Use built-in functions to analyse data.

Steps:

1. Continue with "StudentMarks.xlsx."
2. In B8 type "Sum:" In C8 type `=SUM(B2:B6)`. What result do you get?
3. In B9 type "Average:" In C9 type `=AVERAGE(B2:B6)`.
4. In B10 type "Highest:" In C10 type `=MAX(B2:B6)`.
5. In B11 type "Lowest:" In C11 type `=MIN(B2:B6)`.
6. Repeat for the Science column (column C).
7. Save.

Expected Outcome: Students use 4 functions correctly.

Practical 6: Number Formatting 10 min

You Will Need: Computer 10 min MS Excel

Objective: Apply different number formats to cells.

Steps:

1. Create a new sheet. Enter prices of 5 items in column B.
2. Select B1:B5. Apply **Currency** format (`Ctrl`+`Shift`+`$`).
3. In C1:C5 enter percentages (e.g., 0.85). Apply **Percentage** format.
4. In D1:D5 enter dates. Apply a date format (Format Cells → Date).
5. Compare how the same number looks in different formats.

Expected Outcome: Students apply 3 different number formats.

Practical 7: Creating Charts 15 min

You Will Need: Computer 15 min MS Excel

Objective: Create bar and pie charts from data.

Steps:

1. Open "StudentMarks.xlsx."
2. Select cells A1:C6 (names and marks for both subjects).
3. Insert → Charts → **Clustered Bar Chart**.
4. Add a chart title: "Student Marks Comparison."
5. Now select one student's marks to create a **Pie Chart**.
6. Add data labels to the pie chart (Chart Design → Add Chart Element).
7. Save.

Expected Outcome: A bar chart and a pie chart from the same data.

Practical 8: Sorting and Complete Spreadsheet 15 min

You Will Need: Computer 15 min MS Excel

Objective: Sort data and create a polished spreadsheet.

Steps:

1. Open “StudentMarks.xlsx.”
2. Click anywhere in the Total column. Data tab → Sort Largest to Smallest.
3. Observe how all rows rearrange together.
4. Add conditional formatting: select marks → Home → Conditional Formatting → Colour Scales.
5. Add a merged title row at the top: “Class 8 — Term 1 Results.”
6. Save and check Print Preview.

Expected Outcome: A professional sorted and formatted spreadsheet.

Scan for extra Part V resources and activities!

PART VI

VI

Microsoft PowerPoint

Create stunning presentations that captivate your audience with slides, animations, and transitions.

Slides Animations Media Design Slideshow

Part VI

Microsoft PowerPoint

38 Introduction to MS PowerPoint

Learning Objectives

- Open PowerPoint and identify the main interface elements
- Understand the purpose of presentations and slides
- Create a new presentation and add slides
- Apply themes and design templates

PowerPoint is your tool for telling stories visually. Whether it's a school project, a science presentation, or a birthday slideshow — PowerPoint makes you look like a pro!

38.1 The PowerPoint Window

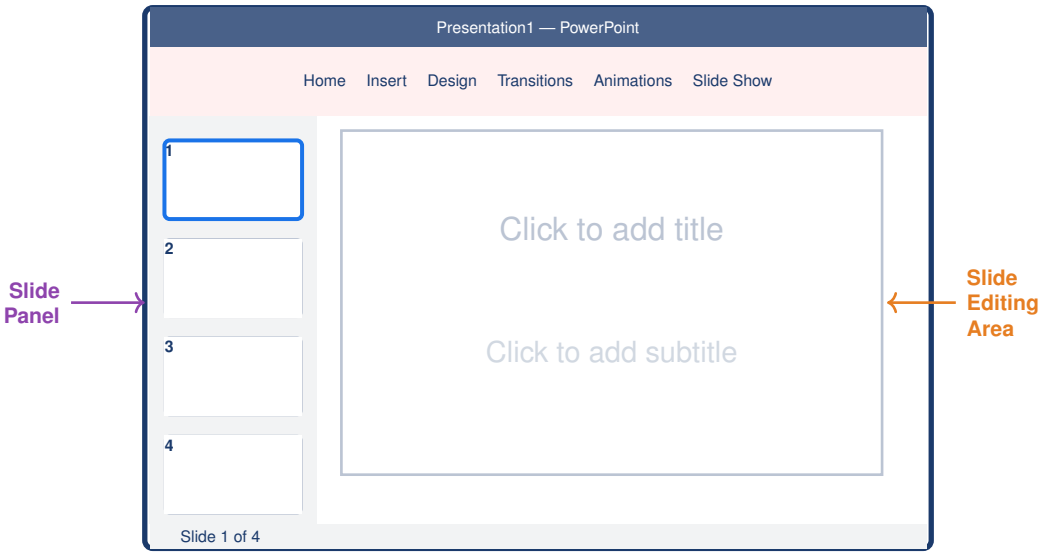


Figure 38.1: The PowerPoint window with key elements labelled

38.2 Creating a Presentation

Creating Your First Presentation

1. Open PowerPoint. Choose a **theme** or "Blank Presentation."
2. Click "Click to add title" and type your title.
3. Click "Click to add subtitle" and type your subtitle.
4. Add a new slide: Home tab → **New Slide** (or `Ctrl`+`M`).
5. Choose a **slide layout** (Title, Content, Comparison, etc.).
6. Save as "MyPresentation.pptx" (`Ctrl`+`S`).

Slide Layouts

PowerPoint offers several built-in slide layouts:

- **Title Slide** — For the first slide (title + subtitle)
- **Title and Content** — Title + bullet points or images
- **Two Content** — Side-by-side content comparison
- **Blank** — Empty slide for custom designs
- **Section Header** — To divide your presentation into parts

The average human attention span is about **8 seconds**. That's why good presentations use visuals, short text, and engaging animations. Nobody wants to read a wall of text on a slide!

Did You Know?

An estimated **35 million PowerPoint presentations** are created every single day worldwide. That's over 400 presentations every second! Microsoft estimates that about 500 million people use PowerPoint regularly.

Review Questions

Q1. Name four parts of the PowerPoint window.

Q2. How do you add a new slide to a presentation?

Q3. Name three slide layouts and describe when you'd use each.

Q4. What is the shortcut to save a presentation?

Q5. What file extension do PowerPoint files use?

Challenge Task

Create a 5-slide presentation about "My Favourite Book/Movie." Slide 1: Title. Slide 2: Summary. Slide 3: Main characters. Slide 4: Why you like it. Slide 5: Rating out of 10. Apply a design theme and save.

39 Designing Slides

Learning Objectives

- Add and format text, images, and shapes on slides
- Apply design themes and customise colours
- Use SmartArt for professional diagrams
- Follow the rules of good slide design

A great slide has **more visuals and less text**. Remember the 6×6 rule: no more than 6 bullet points per slide, and no more than 6 words per bullet!

39.1 Adding Content to Slides

You can add many types of content through the **Insert** tab:

- **Text Boxes** — Add text anywhere on the slide
- **Pictures** — Insert photos from your computer or online
- **Shapes** — Arrows, rectangles, stars, callouts
- **SmartArt** — Professional diagrams (flowcharts, hierarchies, cycles)
- **Tables** — Organised data in rows and columns
- **Videos** — Embed videos directly into slides

39.2 Good Slide Design Rules

✗ Too much text! ✓ Clear and visual!

Figure 39.1: Bad slide (left) vs good slide (right) — less text, more impact

Golden Rules of Slide Design

1. **Keep text short** — Use keywords, not sentences
2. **Use large fonts** — At least size 24 for body text, 36+ for titles
3. **One idea per slide** — Don't overcrowd
4. **Use images and diagrams** — Visuals are remembered 6× better than text
5. **Consistent design** — Use the same theme, fonts, and colours throughout
6. **High contrast** — Dark text on light background, or light text on dark

BYTE

Don't use too many different fonts, colours, or animations in one presentation. It looks chaotic and unprofessional. Stick to **2 fonts**, **3–4 colours**, and **subtle animations**.

Did You Know?

The "Death by PowerPoint" problem is so well-known that Amazon's Jeff Bezos **banned PowerPoint** in meetings! Instead, Amazon employees write memos, and everyone reads them silently at the start of each meeting.

Review Questions

Q1. Name five types of content you can insert into a slide.

Q2. What is SmartArt and when would you use it?

Q3. State three rules of good slide design.

Q4. What is the 6×6 rule?

Q5. Why is high contrast important in slide design?

Challenge Task

Create a 3-slide presentation redesign: take a “bad” slide (with too much text) and improve it by splitting the content across 3 slides with clear visuals, short bullet points, and images. Save as “DesignChallenge.pptx.”

40 Transitions and Animations

Learning Objectives

- Add slide transitions and control their timing
- Apply animations to text, images, and shapes
- Understand the difference between transitions and animations
- Run a slideshow and use presenter tools

Transitions and animations make your presentation come alive! But remember: subtle effects look professional, while excessive effects look childish. Use them wisely!

40.1 Transitions vs Animations

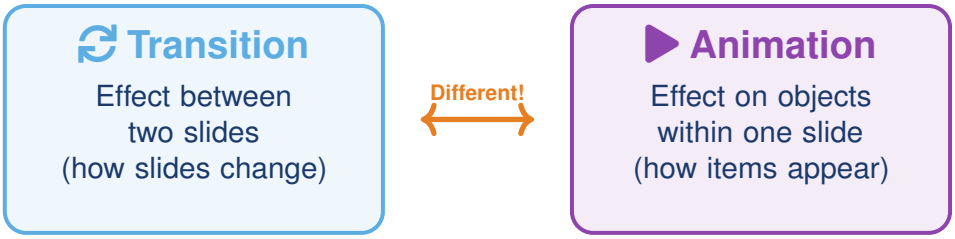


Figure 40.1: Transitions move between slides; animations move objects within a slide

40.2 Adding Transitions

1. Select a slide in the slide panel.
2. Go to the **Transitions** tab.
3. Choose a transition effect (Fade, Push, Wipe, etc.).
4. Set the **duration** (how fast the transition happens).
5. Click “Apply To All” to use the same transition throughout.

40.3 Adding Animations

1. Select the object you want to animate (text, image, or shape).
2. Go to the **Animations** tab.
3. Choose an effect: Appear, Fade In, Fly In, Bounce, etc.
4. Use "Animation Pane" to see and reorder all animations.
5. Set timing: On Click, With Previous, or After Previous.

40.4 Running a Slideshow

Keyboard Shortcuts

F5	Start slideshow from the beginning
Shift + F5	Start from the current slide
Esc	Exit the slideshow
B	Black screen (during show)
W	White screen (during show)

BYTE

The "B" and "W" keys during a slideshow are secret presentation tricks! Press B to black out the screen when you want the audience to focus on you, not the slides. Press W for a white screen. Press any key to return.

Did You Know?

PowerPoint was originally created by Robert Gaskins and Dennis Austin in 1987 for Macintosh, and was called **"Presenter."** Microsoft bought the company (Forethought) for $14 million just 3 months after its release — one of the best investments in tech history!

Review Questions

Q1. What is the difference between a transition and an animation?

Q2. How do you add a transition to a slide?

Q3. Name three animation effects available in PowerPoint.

Q4. What keyboard shortcut starts a slideshow from the beginning?

Q5. Why should you avoid using too many animation effects?

Challenge Task

Take your “Favourite Book/Movie” presentation from Chapter 33. Add a different transition to each slide. Animate the title on slide 1 with “Fly In,” and animate bullet points on slide 2 to appear one by one. Run the slideshow for a partner!

41 Part VI — Hands-On Practicals

BYTE

It's showtime! These practicals will turn you into a presentation wizard. Lights, slides, action!

Practical 1: Creating Your First Presentation 15 min

You Will Need: Computer 15 min MS PowerPoint

Objective: Create a basic 5-slide presentation.

Steps:

1. Open PowerPoint and choose a theme.
2. Slide 1: Add a title "About Me" and your name as subtitle.
3. Add 4 more slides (Home → New Slide): My Family, My Hobbies, My School, Thank You.
4. Add bullet points on each content slide.
5. Save as "AboutMe.pptx."
6. Run the slideshow (F5) and click through all slides.

Expected Outcome: A 5-slide presentation with consistent theme.

Practical 2: Inserting Images and Shapes 15 min

You Will Need: Computer 15 min MS PowerPoint

Objective: Enhance slides with visual elements.

Steps:

1. Open "AboutMe.pptx."
2. On "My Hobbies" slide, insert a picture (Insert → Pictures).
3. Resize the image and position it next to your text.
4. Insert a shape (star, arrow, or callout) and add text inside.
5. On another slide, insert SmartArt (Insert → SmartArt) to show a process.
6. Save.

Expected Outcome: Slides with images, shapes, and SmartArt.

Practical 3: Applying Transitions 10 min

You Will Need: Computer 10 min MS PowerPoint

Objective: Add transitions between slides.

Steps:

1. Open "AboutMe.pptx."

2. Click on Slide 1. Go to Transitions tab. Choose “Fade.”
3. Click on Slide 2 and choose “Push.” Set duration to 1 second.
4. Apply different transitions to remaining slides.
5. Click “Apply To All” to test — then undo and keep individual transitions.
6. Run slideshow to see the transitions in action.

Expected Outcome: Smooth transitions between all slides.

Practical 4: Adding Animations 15 min

You Will Need: Computer 15 min MS PowerPoint

Objective: Animate text and objects within slides.

Steps:

1. On Slide 1, select the title. Animations tab → “Fly In.”
2. Select the subtitle. Apply “Fade” animation.
3. On a content slide, select the bullet text. Apply “Appear” animation.
4. Click “Effect Options” → “By Paragraph” so bullets appear one by one.
5. Select a picture. Apply “Grow & Turn” animation.
6. Open Animation Pane to review the order and timing.

Expected Outcome: Objects animate in sequence within slides.

Practical 5: Presentation on a Topic 20 min

You Will Need: Computer 20 min MS PowerPoint

Objective: Create a complete, polished presentation on a given topic.

Steps:

1. Choose a topic: “Saving the Environment” or “Space Exploration.”
2. Create 8 slides: Title, Introduction, 4 Content slides, Summary, Thank You.
3. Use at least: 2 images, 1 SmartArt, 1 table, consistent transitions.
4. Apply the 6×6 rule (max 6 bullets, max 6 words each).
5. Add speaker notes at the bottom of each slide (View → Notes).
6. Save and present to a partner using slideshow mode.

Expected Outcome: A professional 8-slide presentation with all elements.

Practical 6: Slideshow Presentation Skills 10 min

You Will Need: Computer 10 min MS PowerPoint

Objective: Present confidently using slideshow tools.

Steps:

1. Open your presentation. Press `F5` to start.
2. Practise advancing slides with clicks and arrow keys.
3. Press `B` to black out the screen. Press any key to return.
4. Press `Shift` + `F5` to start from the current slide.
5. Use the pointer tool (right-click → Pointer Options → Laser Pointer).
6. Present to a partner — speak clearly! Limit slides to 30 seconds each.

Expected Outcome: Students present confidently using slideshow controls.

Scan for extra Part VI resources and activities!

PART VII

The Internet and the Web

Explore the world wide web, search like a pro, stay safe online, and use email and cloud tools.

Web Search Email Safety Cloud

Part VII

The Internet and the Web

42 How the Internet Works

Learning Objectives

- Define the Internet and explain how it connects people
- Understand the roles of ISPs, routers, and servers
- Differentiate between the Internet and the World Wide Web
- Know what URLs, links, and web pages are

The Internet is the biggest network ever created — it connects **billions** of devices across every country on Earth. Let's understand how this magic works!

42.1 What Is the Internet?

Definition: The Internet

The **Internet** is a global network of interconnected computers that communicate using standard protocols (rules). It allows people to share information, communicate, and access services from anywhere in the world.

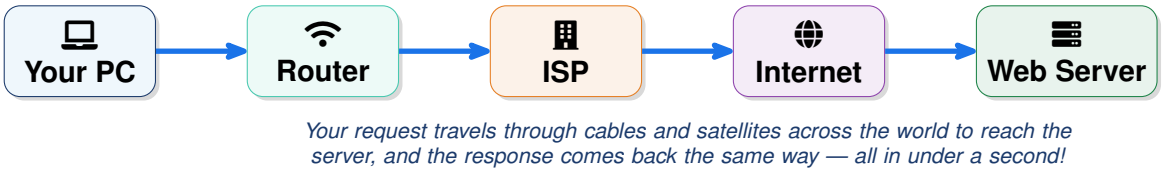


Figure 42.1: How your computer connects to a website on the Internet

42.2 Internet vs World Wide Web

- The **Internet** is the *physical network* — the cables, routers, and satellites
- The **World Wide Web** (WWW) is a *service* that runs on the Internet — the collection of websites you browse
- Think of the Internet as the **road system**, and the Web as the **shops and buildings** along those roads

42.3 Understanding URLs

Figure 42.2: Anatomy of a URL — every part has a purpose

The first-ever website was created by **Tim Berners-Lee** on December 20, 1990. It described the World Wide Web project itself. The website is still online today at `info.cern.ch`!

Did You Know?

There are currently over **2 billion websites** on the Internet, but only about 400 million are actively maintained. The rest are "parked" domain names or abandoned sites.

Review Questions

Q1. What is the Internet? Define it in your own words.

Q2. What is the difference between the Internet and the World Wide Web?

Q3. What does ISP stand for, and what does it do?

Q4. What are the parts of a URL?

Q5. Who invented the World Wide Web?

Challenge Task

Visit **5 different websites** and write down the full URL of each. For each URL, identify: the protocol, domain name, and path. Present your findings in a neat table.

43 Web Browsing and Searching

Learning Objectives

- Use a web browser to navigate websites
- Search effectively using Google
- Bookmark favourite websites
- Evaluate website reliability and trustworthiness

BYTE

Searching the web is a skill — not all results are reliable, and knowing the right keywords makes you **10× faster** at finding what you need!

43.1 Using a Web Browser

A web browser is an application that lets you view websites. Popular browsers include **Google Chrome**, **Microsoft Edge**, **Mozilla Firefox**, and **Safari**.

43.2 Searching Effectively

Smart Search Tips

1. **Use keywords**, not full sentences — "Indian Independence Day date" instead of "When is Indian Independence Day?"
2. **Use quotes** for exact phrases — `"solar system planets"`
3. **Add** `site:` to search within one site — `site:wikipedia.org dinosaurs`
4. **Use minus** to exclude words — `jaguar -car` (to find the animal)
5. **Check multiple sources** — don't trust just one website

43.3 Evaluating Websites

Not everything on the internet is true! Use the **CRAAP test** to check reliability:

Table 43.1: The CRAAP test for evaluating websites

Letter	Stands For	Ask Yourself
C	Currency	Is the information up to date?
R	Relevance	Does it answer your question?
A	Authority	Who wrote it? Are they an expert?
A	Accuracy	Can you verify the facts elsewhere?
P	Purpose	Why was it written? To inform, sell, or persuade?

Anyone can publish anything on the internet. Just because it looks professional doesn't mean it's true. Always verify important facts with at least **two reliable sources**!

Did You Know?

Google processes over **8.5 billion searches per day**. That's about 99,000 searches every single second! The most Googled question ever is "What to watch?"

Review Questions

Q1. Name three popular web browsers.

Q2. What is a search engine and name two examples.

Q3. List three tips for searching effectively on Google.

Q4. What is the CRAAP test?

Q5. Why should you check multiple sources for information?

Challenge Task

Search for the answer to this question: "How tall is the Eiffel Tower in metres?" Use **three different websites** to find the answer. Do they all agree? Write down each source and its answer. Which source do you trust most, and why?

44 Email and Online Communicatio

Learning Objectives

- Create and send emails with proper etiquette
- Attach files to an email
- Understand CC, BCC, and Reply All
- Recognise phishing emails and online scams

Email is one of the most important communication tools in the world. Learning to write proper emails now will help you throughout school, college, and your career!

44.1 What Is Email?

Email (Electronic Mail) is a way to send and receive digital messages over the Internet. Popular services include **Gmail**, **Outlook**, and **Yahoo Mail**.

44.2 Parts of an Email

Parts of an Email

To	The recipient's email address
CC	Carbon Copy — send a copy to others (visible to all)
BCC	Blind Carbon Copy — copy to others (hidden from all)
Subject	A short title describing the email's purpose
Body	The main message content
Attachments	Files sent along with the email
Signature	Your name and contact info at the bottom

44.3 Email Etiquette

1. Always write a clear, specific **subject line**
2. Start with a **greeting** (Dear Sir/Ma'am, Hello, Hi)
3. Keep the message **short and clear**

4. Use **polite language** — no ALL CAPS (it looks like shouting!)
5. Always **proofread** before sending
6. End with a **sign-off** (Kind regards, Thank you, Best wishes)

Warning!

Phishing alert! Watch out for emails that:

- Ask for your **password** or personal information
- Contain **urgent threats** ("Your account will be deleted!")
- Come from **unknown senders** with suspicious links
- Have **spelling mistakes** in the sender's address (e.g., "gooogle.com")

When in doubt, **don't click any links** — ask a parent or teacher first!

Once you send an email, you **can't un-send it** (well, Gmail gives you about 30 seconds). Always double-check the recipient and re-read your message before hitting Send!

Did You Know?

The first email was sent in 1971 by **Ray Tomlinson**. He chose the @ symbol to separate the user name from the computer name. He later admitted he couldn't remember what the first email said — probably "QWERTYUIOP" or something similar as a test!

Review Questions

Q1. What does CC stand for, and how is it different from BCC?

Q2. Name three things you should include in a formal email.

Q3. What are three signs that an email might be a phishing scam?

Q4. Why should you always write a clear subject line?

Q5. How do you attach a file to an email?

Challenge Task

Write a **formal email** to your teacher asking for extra time on a homework assignment. Include: proper greeting, clear subject line, polite body with a reason, and a professional sign-off. Save it as a draft or send it for real!

45 Online Safety and Digital Citizen

Learning Objectives

- Understand the concept of digital citizenship
- Identify online threats: cyberbullying, scams, inappropriate content
- Protect personal information online
- Follow rules for responsible internet use

BYTE

The internet is amazing, but it can also be dangerous if you're not careful. Being a good **digital citizen** means being safe, kind, and smart online.

45.1 What Is Digital Citizenship?

Digital Citizenship

Digital Citizenship means using technology responsibly, ethically, and safely. A good digital citizen:

- ✓ Treats others with **respect** online
- ✓ Protects their **personal information**
- ✓ Thinks before they **post or share**
- ✓ Reports **bullying or harmful content**
- ✓ Uses technology to **learn and create**, not harm

45.2 Staying Safe Online

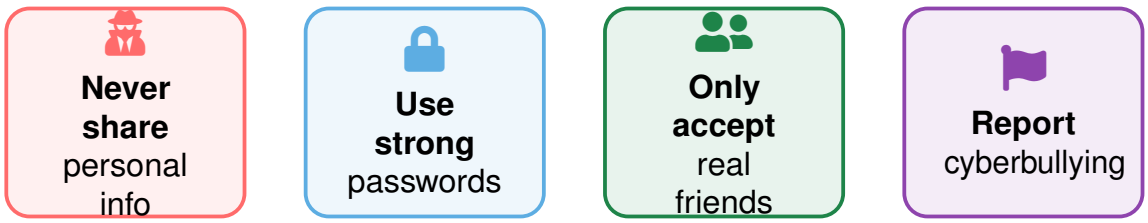


Figure 45.1: Four golden rules for staying safe online

45.3 Cyberbullying

Warning!

Cyberbullying is using technology to harass, threaten, or humiliate someone. This includes:

- Sending mean messages or comments
- Sharing someone's photos without permission
- Excluding someone from online groups to hurt them
- Creating fake profiles to impersonate someone

If you see cyberbullying: **don't respond**, **save evidence**, and **tell a trusted adult immediately**.

The **Golden Rule** of the internet: Don't say or do anything online that you wouldn't say or do face-to-face. If you wouldn't show it to your grandmother, don't post it!

Did You Know?

In many countries, cyberbullying is a **criminal offence**. In India, Section 67 of the IT Act can lead to up to **3 years in jail** for sharing offensive or harmful content online. The internet is NOT a lawless place!

Review Questions

Q1. What is digital citizenship? List three qualities of a good digital citizen.

Q2. Name four types of personal information you should never share online.

Q3. What is cyberbullying, and how should you respond if you see it?

Q4. Why should you think twice before posting something online?

Q5. What are three rules for staying safe on social media?

Challenge Task

Create a "Digital Safety Pledge" poster with at least 8 rules for safe internet use. Design it colourfully and hang it near your computer at home or school!

46 Part VII — Hands-On Practicals

Let's go online — safely! These practicals teach you to browse, search, email, and stay protected while exploring the digital world.

Practical 1: Exploring a Web Browser 10 min

You Will Need: Computer with Internet 10 min Browser

Objective: Navigate a web browser and identify its parts.

Steps:

1. Open Google Chrome (or your school's default browser).
2. Identify: Address Bar, Tabs, Refresh Button, Back/Forward Buttons.
3. Type `www.wikipedia.org` in the address bar and press Enter.
4. Click on any link to navigate to a new page. Use the Back button.
5. Open a new tab (Ctrl+T) and visit `www.google.com`.

Expected Outcome: Students navigate browser elements confidently.

Practical 2: Smart Searching 15 min

You Will Need: Computer with Internet 15 min Google

Objective: Use advanced search techniques.

Steps:

1. Search: “planets in the solar system” — note the results.
2. Search with quotes: `"largest planet in the solar system"` — compare results.
3. Search with `site:`: `site:wikipedia.org Indian festivals`.
4. Search with minus: `python -snake` (to find the programming language).
5. For each search, write down how many results appeared and the top result.

Expected Outcome: Students use 4 search techniques effectively.

Practical 3: Bookmarking Websites 10 min

You Will Need: Computer with Internet 10 min Browser

Objective: Save and organise favourite websites.

Steps:

1. Visit 3 educational websites (e.g., Wikipedia, National Geographic Kids, Khan Academy).

2. Bookmark each one: click the star icon in the address bar.
3. Create a bookmark folder called “School Resources.”
4. Move all 3 bookmarks into that folder.
5. Access a bookmarked site from the bookmarks bar.

Expected Outcome: Students save and organise 3 bookmarks.

Practical 4: Composing an Email 15 min

You Will Need: Computer with Internet | 15 min | Email account

Objective: Write and send a properly formatted email.

Steps:

1. Open your email (Gmail, Outlook, or school email).
2. Click “Compose” (or “New Email”).
3. In the “To” field, type your teacher’s email address.
4. Subject: “Homework Query — [Your Name].”
5. Body: Write a polite message asking about a homework deadline.
6. Add your name as a sign-off. Click Send (or save as Draft).

Expected Outcome: Students compose a well-structured email.

Practical 5: Identifying Phishing Emails 10 min

You Will Need: Computer 10 min Worksheet

Objective: Spot the signs of phishing emails.

Steps:

1. Your teacher will show you 5 sample emails (printed or on screen).
2. For each email, decide: REAL or PHISHING?
3. Write down the clues that helped you decide.
4. Discuss your answers with the class.
5. Create a “Phishing Red Flags” checklist with at least 5 warning signs.

Expected Outcome: Students correctly identify 4 out of 5 phishing emails.

Practical 6: Digital Citizenship Pledge 15 min

You Will Need: Computer 15 min MS Word or Paper

Objective: Create a personal digital citizenship pledge.

Steps:

1. Open MS Word (or use paper).
2. Write a title: “My Digital Citizenship Pledge.”
3. Write 8 promises about safe, kind, and responsible internet use.
4. Add your name, date, and signature at the bottom.
5. Decorate it and print it (or display it on screen).
6. Share your pledge with the class.

Expected Outcome: A personalised digital citizenship pledge document.

Scan for extra Part VII resources and activities!

PART VIII

Review, Projects and Beyond

Bring everything together with cross-module projects, revision exercises, and a glimpse into the future of technology.

Projects Revision Innovation Future Tech Challenges

Part VIII

Review, Projects, and Beyond

47 Revision — Computer Fundame

Learning Objectives

- Revise key concepts from Parts I–III
- Test your knowledge with comprehensive review questions
- Reinforce understanding through diagrams and tables
- Identify areas that need more practice

This is your one-stop revision chapter for everything we covered about computer basics, hardware, software, and Windows. Use it to prepare for tests!

47.1 Key Concepts Summary

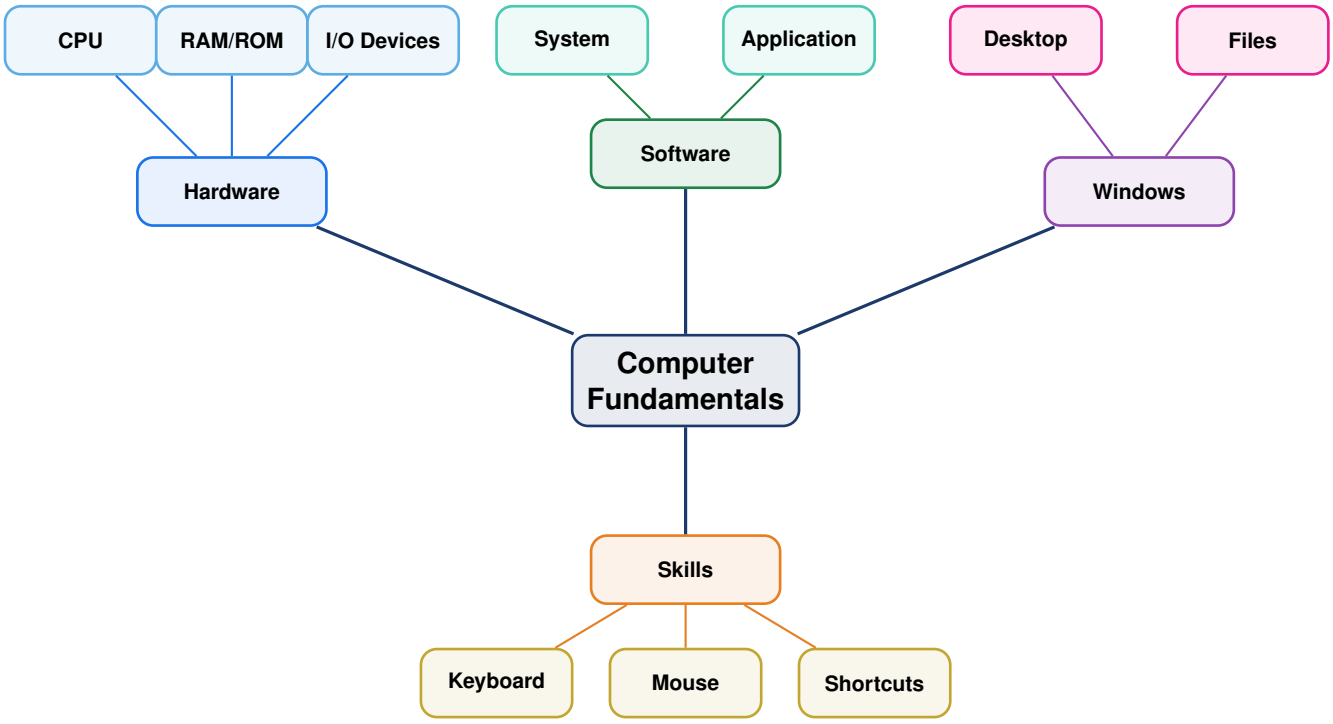


Figure 47.1: Master concept map for computer fundamentals

47.2 Quick-Fire Revision

Table 47.1: Quick-fire revision — fill in the blanks

#	Statement	Answer
1	CPU stands for _____	
2	RAM is _____ memory (volatile/non-volatile)	
3	The two main types of software are _____ and _____	
4	The shortcut to copy is _____	
5	A file's address in Explorer is called its _____	
6	The Recycle Bin stores _____ files	
7	GUI stands for _____	
8	POST stands for _____	
9	_____ is the brain of the computer	
10	_____ key + S saves a file	

Review Questions

Q1. Draw and label the IPO cycle.

Q2. List the 5 characteristics of a computer.

Q3. Compare RAM and ROM in a table with at least 4 differences.

Q4. Explain the boot process in 5 steps.

Q5. Name 5 input devices and 5 output devices.

Q6. What is the difference between hardware and software?

Q7. Write the keyboard shortcuts for: Copy, Cut, Paste, Undo, Save, Select All.

Q8. What are the main elements of the Windows desktop?

Q9. Explain the file naming rules in Windows.

Q10. Draw a folder hierarchy with at least 3 levels.

48 Revision — MS Office and Intern

Learning Objectives

- Revise key concepts from Parts IV–VII (Word, Excel, PowerPoint, Internet)
- Compare features across MS Office applications
- Test your knowledge with practical review tasks
- Consolidate internet safety and email skills

This is Part 2 of your revision — covering all your MS Office skills and internet knowledge. If you can answer everything here, you're ready for any test!

48.1 MS Office Quick Comparison

Table 48.1: Comparing the three MS Office applications

Feature	MS Word	MS Excel	MS PowerPoint
Purpose	Documents	Data & calculations	Presentations
Extension	.docx	.xlsx	.pptx
Main area	Page	Worksheet (cells)	Slide
Key feature	Text formatting	Formulas & charts	Animations
Best for	Reports, letters	Budgets, marks	Slideshows

48.2 MS Word Revision

Review Questions

Q1. Name 6 parts of the MS Word window.

Q2. What is the difference between Save (Ctrl+S) and Save As (F12)?

Q3. List 4 font formatting options and their shortcuts.

Q4. How do you insert a table, picture, and page number in Word?

Q5. What is the difference between Portrait and Landscape orientation?

48.3 MS Excel Revision

Review Questions

Q1. What is a cell address? Give an example.

Q2. Write the formulas for: SUM of B1 to B10, AVERAGE of C1 to C5.

Q3. What does AutoFill do, and how do you use it?

Q4. Name three chart types and when to use each.

Q5. How do you sort data in ascending order?

48.4 MS PowerPoint Revision

Review Questions

Q1. What is the difference between transitions and animations?

Q2. Name 4 slide layouts in PowerPoint.

Q3. What is the 6×6 rule for slide design?

Q4. What keyboard shortcut starts a slideshow?

Q5. List 3 golden rules of good slide design.

48.5 Internet & Safety Revision

Review Questions

Q1. What is the difference between the Internet and the World Wide Web?

Q2. Who invented the WWW?

Q3. What are the parts of a URL?

Q4. Name 4 signs of a phishing email.

Q5. What is digital citizenship? List 3 responsibilities.

Q6. Explain the CRAAP test for evaluating websites.

Challenge Task

Create a **one-page cheat sheet** (in Word or by hand) summarising the most important keyboard shortcuts, definitions, and concepts from the entire book. Make it colourful, organised, and something you'd want to stick on your wall!

49 Cross-Module Projects

Learning Objectives

- Apply skills from multiple modules in real-world projects
- Work independently on extended assignments
- Create professional-quality documents, spreadsheets, and presentations
- Present your work to the class

These projects combine EVERYTHING you've learned. Think of them as your "final boss" challenges! Complete them and you'll be a true computer whiz.

49.1 Project 1: My Digital Portfolio

Project Brief

Create a **digital portfolio** showcasing your best work from this course.
Deliverables:

1. **Word Document:** A 2-page "About Me" profile with formatted text, a table of skills, an inserted photo, headers/footers, and page numbers. Save as PDF.
2. **Excel Spreadsheet:** A "Skills Tracker" with your self-rated scores (1–10) for 8 skills. Include SUM, AVERAGE formulas, and a bar chart showing your strengths.
3. **PowerPoint Presentation:** A 6-slide "My Journey in Computer Science" presentation with transitions, animations, and at least one embedded chart or SmartArt.
4. **Folder Structure:** Organise all files in a folder called "MyPortfolio" with sub-folders: Documents, Spreadsheets, Presentations.

Time: 2–3 class periods
Assessment: Presentation quality, proper formatting, organisation, creativity.

49.2 Project 2: School Event Planner

>_ Project Brief

Plan a school event (Sports Day, Science Fair, or Cultural Festival) using MS Office.
Deliverables:

1. **Word:** An invitation letter to parents with WordArt title, formatted paragraphs, and event details in a table.
2. **Excel:** A budget spreadsheet with items, estimated costs, actual costs, and formulas for totals and difference. Include a pie chart of spending categories.
3. **PowerPoint:** A 10-slide presentation to pitch the event to your principal. Include timeline (SmartArt), budget summary (chart), and team roles.
4. **Email:** Draft a formal email to a teacher inviting them to be a judge, with the invitation letter attached.

Time: 3–4 class periods
Assessment: Completeness, professionalism, effective use of all three Office apps.

49.3 Project 3: Technology Research Report

>_ Project Brief

Research an emerging technology and create a comprehensive report.
Choose one topic:

- Artificial Intelligence (AI)
- Virtual Reality (VR)
- 3D Printing
- Self-Driving Cars
- Space Technology

Deliverables:

1. **Word:** A 3-page research report with: title page, introduction, 3 main sections, conclusion, and a bibliography of at least 3 sources. Use proper formatting, headings, and at least one image.
2. **Excel:** A data table with at least 10 facts/statistics about your topic. Create a chart from the data.
3. **PowerPoint:** A 8-slide presentation summarising your research for the class.

Time: 3–4 class periods
Assessment: Research quality, source reliability (CRAAP test), visual design, presentation skills.

For all projects: plan before you start, save frequently (Ctrl+S!), and proofread everything. A little extra effort on presentation makes a **huge** difference in how your work is received.

Did You Know?

Many technology companies like Google, Apple, and Microsoft still use presentations, spreadsheets, and documents as their primary tools for business communication. The skills you're learning right now are the *exact same skills* used in offices around the world!

50 The Future of Technology

Learning Objectives

- Explore emerging technologies shaping the future
- Understand the basics of AI, IoT, and cloud computing
- Think critically about technology's impact on society
- Get inspired to continue learning about computer science

You've reached the final chapter — congratulations! Now let's look at what's coming next in the world of technology. Spoiler: it's AMAZING.

50.1 Emerging Technologies

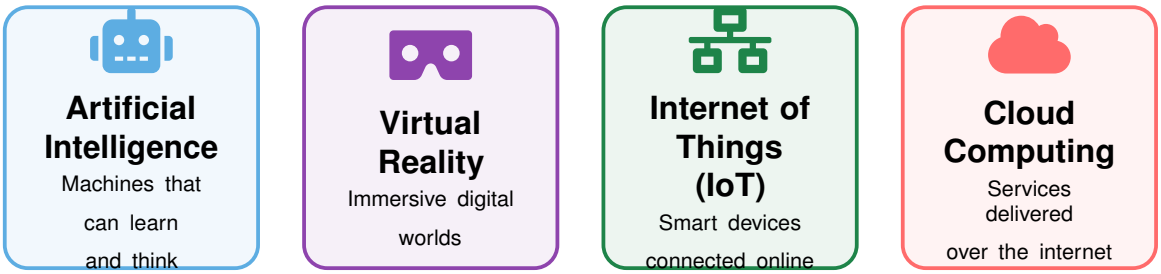


Figure 50.1: Four technologies shaping our future

50.2 Artificial Intelligence (AI)

Artificial Intelligence is technology that enables machines to learn from data, recognise patterns, and make decisions — almost like a human brain, but digital.

Real World Connection

You already use AI every day! When YouTube recommends videos, when your phone recognises your face, when Google completes your search query, and when Siri or Alexa answers your questions — that's all AI at work!

50.3 Internet of Things (IoT)

The Internet of Things connects everyday objects to the internet — from smart watches and fitness trackers to smart fridges and home assistants.

Table 50.1: Examples of IoT devices

Device	What It Does	How It Helps
Smart Watch	Tracks health, shows notifications	Monitors heart rate, steps
Smart Speaker	Plays music, answers questions	Voice-controlled assistance
Smart Thermostat	Controls home temperature	Saves energy automatically
Smart Lock	Locks/unlocks doors remotely	Security from your phone

50.4 Your Journey Continues!

This book has given you a strong foundation in computer science. But this is just the **beginning**! Here are some exciting paths you can explore next:

- **Programming** — Learn Python, Scratch, or JavaScript to create your own programs
- **Web Development** — Build your own websites with HTML, CSS, and JavaScript
- **Robotics** — Build and program real robots
- **Cybersecurity** — Protect systems from hackers and threats
- **Game Development** — Create your own video games
- **Data Science** — Analyse data to discover patterns and insights

The most important skill in computer science isn't memorising facts — it's **curiosity**. Keep asking "How does this work?" and "What if I tried this?" That's how every great inventor started!

Did You Know?

The term "Artificial Intelligence" was coined in 1956 by **John McCarthy** at a conference at Dartmouth College. He predicted that machines would be as smart as humans within 20 years. It took a bit longer than that, but we're getting closer every day!

Review Questions

Q1. What is Artificial Intelligence? Give two examples of AI you use daily.

Q2. What does IoT stand for? Name three IoT devices.

Q3. What is cloud computing, and name two cloud services.

Q4. How might AI change education in the next 10 years?

Q5. What area of computer science interests you most, and why?

Challenge Task

Write a **one-page essay** titled "The Computer of 2050" describing what you think computers will look like and be able to do in 25 years. Will they be wearable? Will they read your mind? Use your imagination — and back it up with real technology trends!

Congratulations!

You've completed **Foundations of Computer Science**!

You now have the skills to use computers confidently, create professional documents, and stay safe online.

Keep exploring. Keep creating. Keep learning!

— Byte the Robot

50 Glossary

ALU (Arithmetic Logic Unit)
The part of the CPU that performs all arithmetic (maths) and logical (comparison) operations.

Application Software
Programs designed for end-users to perform specific tasks, such as MS Word, Chrome, or Calculator.

AutoFill An Excel feature that automatically fills cells with a series or pattern based on initial values.

Backup A copy of data stored separately to protect against data loss.

Boot Process
The sequence of events that occurs when a computer is turned on, from POST to the desktop appearing.

Browser An application used to view websites on the Internet (e.g., Chrome, Edge, Firefox).

Cell Address
The unique identifier of a cell in Excel, made of its column letter and row number (e.g., B3).

Cloud Storage
Online storage that allows files to be saved and accessed over the Internet (e.g., Google Drive, OneDrive).

CPU (Central Processing Unit)
The “brain” of the computer that processes instructions and manages data.

Cyberbullying
Using technology to harass, threaten, or humiliate someone online.

Desktop The main screen of the Windows operating system where icons, taskbar, and windows appear.

Device Driver
A small program that allows the operating system to communicate with a specific hardware device.

Digital Citizenship
Responsible, ethical, and safe use of technology and the Internet.

Email Electronic mail; a method of sending digital messages over the Internet.

Ethernet A wired network connection using a cable plugged into a router or network port.

Extension (File)
The suffix after the dot in a filename that indicates the file type (e.g., .docx, .pdf, .jpg).

File A collection of data stored with a name and extension on a storage device.

Firmware
Permanent software built into hardware devices (e.g., BIOS on a motherboard).

Folder A container used to organise files on a storage device; also called a directory.

Formula An expression in Excel that performs calculations using cell references, operators, and functions.

Function (Excel)
A pre-built formula in Excel for common calculations (e.g., SUM, AVERAGE, MAX, MIN).

GUI (Graphical User Interface)
A visual interface using windows, icons, menus, and pointers to interact with the computer.

Hardware
The physical components of a computer that you can touch (e.g., monitor, keyboard, CPU).

Internet A global network of interconnected computers that communicate using standard protocols.

IoT (Internet of Things)
The network of everyday physical devices connected to the Internet, capable of collecting and sharing data.

ISP (Internet Service Provider)
A company that provides Internet access to homes, schools, and businesses.

Keyboard Shortcut
A combination of keys pressed together to perform a quick action (e.g., Ctrl+C to copy).

Malware Malicious software designed to damage, disrupt, or gain unauthorised access to a computer.

Operating System (OS)
System software that manages hardware, provides a user interface, and runs application software.

Phishing A type of online scam using fake emails or websites to trick users into sharing personal information.

POST (Power-On Self-Test)
A diagnostic test run by the computer at startup to check that essential hardware is working.

RAM (Random Access Memory)
Volatile memory used to store data temporarily while programs are running.

Ribbon The toolbar area in MS Office applications containing tabs, groups, and commands.

ROM (Read-Only Memory)
Non-volatile memory containing permanent instructions (e.g., BIOS firmware).

Search Engine
A tool that helps find information on the Internet (e.g., Google, Bing, Yahoo).

Slide (PowerPoint)
A single page in a PowerPoint presentation that displays content.

Software A set of instructions (programs) that tells the computer hardware what to do.

Spreadsheet
A document organised in rows and columns of cells, used for data, calculations, and charts.

System Software
Software that manages hardware and provides a platform for other software (e.g., Windows, Linux).

Taskbar The bar at the bottom of the Windows desktop showing open programs, pinned apps, and system tray.

Transition (PowerPoint)
A visual effect that plays when moving from one slide to the next.

URL (Uniform Resource Locator)
The web address used to locate a specific page on the Internet (e.g., https://www.google.com).

USB (Universal Serial Bus)
A standard connector for attaching devices (flash drives, keyboards, etc.) to a computer.

Virus A type of malware that spreads by infecting other files and can damage the computer system.

Wi-Fi A wireless networking technology that allows devices to connect to the Internet without cables.

World Wide Web (WWW)
A collection of websites and web pages accessible via the Internet using a browser.

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